



Monographs of the Western North American Naturalist

Volume 15 – Issue in Progress

Article 1

5-15-2024

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Recommended Citation

Marsh, Paul C.; Dowling, Thomas E.; Turner, Thomas F.; Osborne, Megan J.; and Kesner, Brian R. (2024)

"Maturation of an off-channel habitat concept to conserve native fishes in the Lower Colorado River,"

Monographs of the Western North American Naturalist. Vol. 15, Article 1.

Available at: <https://scholarsarchive.byu.edu/mwnan/vol15/iss1/1>

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Maturation of an off-channel habitat concept to conserve native fishes in the Lower Colorado River

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ABSTRACT.—Two endemic, “large river” fishes of the Colorado River basin of western North America, bonytail *Gila elegans* and razorback sucker *Xyrauchen texanus*, are among several critically endangered species in the system. Wild populations of bonytail are gone, and there are no self-sustaining populations of razorback sucker anywhere; reproduction occurs but recruitment does not. Both species have been under intensive management in the Lower Colorado River since the 1980s. Today, with the single exception of Lake Mead, remaining populations are composed entirely of repatriated individuals that depend on stocking for their continued existence. In 2003, a conceptual off-channel habitat (OCH) management plan for these and other large river fishes of the system was published by the late W.L. Minckley and colleagues. The cornerstone of the approach was to move away from hatchery-based fish production and instead use isolated OCHs that were free of predatory and competitive nonnative fishes, where populations of native species could live, grow, reproduce, and recruit. Populations of adult fish also would live in open waters of the system, and through active management, individuals would be exchanged with those in OCHs to maintain genetic integrity and diversity of both species. Progress in the last 2 decades toward implementing the plan includes creation of new OCHs, studies of population dynamics and genetics of “wild” and captive populations, development of appropriate metrics to assess status of OCH populations, and refinement of the OCH concept itself. Our goals in this paper are to review management of bonytail and razorback sucker in the Lower Colorado River, present examples of species dynamics in OCHs, offer data-driven refinements to the OCH concept, and explore practical aspects including challenges and constraints to implementation of the concept. We conclude that bonytail and razorback sucker never can meet quantitative criteria required by current recovery plans, but long-term conservation of these species can be achieved if an OCH concept of management is successfully implemented and maintained.

RESUMEN.—Dos peces grandes de río, endémicos de la cuenca del río Colorado en el oeste de América del Norte, el pez carpa elegante *Gila elegans* y el pez matalote jorobado *Xyrauchen texanus*, se encuentran entre varias especies en peligro crítico de extinción en el sistema. Las poblaciones silvestres de *G. elegans* han desaparecido y no hay poblaciones autosostenibles de *X. texanus* en ninguna parte; si bien ocurre la reproducción, no hay reclutamiento. Ambas especies han estado bajo gestión intensiva en la parte baja del río Colorado desde la década de 1980. Hoy en día, con la única excepción del lago Mead, las poblaciones restantes están compuestas en su totalidad por individuos repatriados que dependen del reabastecimiento para continuar existiendo. En 2003, W.L. Minckley (finito) y colegas publicaron un plan conceptual de gestión del hábitat fuera del canal (OCH, por sus siglas en inglés) para estos y otros grandes peces fluviales del sistema. El enfoque principal de la publicación fue evitar la producción de peces en criaderos y, en su lugar, utilizar los del OCH aislados que estuvieran libres de peces no nativos depredadores y competitivos, donde las poblaciones de especies nativas pudieran vivir, crecer, reproducirse y ser reclutados. Las poblaciones de peces adultos también vivirían en aguas abiertas del sistema y, mediante una gestión activa, los individuos se intercambiarían con los de los del OCH para mantener la integridad genética y la diversidad de ambas especies. El progreso en las últimas dos décadas hacia la implementación del plan incluye la creación de nuevos OCH, estudios de dinámica poblacional y genética de poblaciones “silvestres” y cautivas, el desarrollo de métricas apropiadas para evaluar el estado de las poblaciones del OCH y el refinamiento del concepto mismo. Nuestros objetivos en este documento son: analizar la gestión de ambos tipos de peces en la parte baja del río Colorado, presentar ejemplos de dinámicas de especies en el OCH, ofrecer mejoras basadas en datos al concepto del OCH y explorar aspectos prácticos que incluyen los desafíos y las limitaciones para la implementación del concepto. Concluimos que, ninguno de estos peces podría cumplir con los criterios cuantitativos necesarios de los planes de recuperación actuales, pero se podría lograr la conservación de estas especies a largo plazo si se implementa y mantiene exitosamente un concepto de gestión de OCH.

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Anthropogenic changes have significantly influenced freshwater ecosystems globally with negative impacts to biodiversity and our ability to conserve it (Dudgeon et al. 2006, Reid et al. 2019). Among 1187 described species of freshwater and diadromous fishes of North America, 700 extant taxa are imperiled (230 vulnerable, 190 threatened, and 280 endangered) and 61 species are extinct or extirpated from nature (Jelks et al. 2008). Recent loss of diversity has been especially dramatic among fishes of the desert southwest and the Colorado River basin. Environmental and geological factors in this region led to isolation and high levels of endemism (Smith et al. 2010) but reduced levels of biodiversity. The Colorado River basin is home to 40 freshwater species in 20 genera and 9 families (Carlson and Muth 1989); 3 species are extinct, at least 22 (55%) of the remainder are federally protected as threatened or endangered, and others are under consideration for such status. The proportion of at-risk taxa is even higher in some subbasins (e.g., Gila River basin of Arizona and New Mexico). Habitat loss and modification plus introduction and establishment of nonnative species are the most often cited explanations of the imperiled status of this fauna, compounded by myriad other reasons. Competitive and predatory nonnative fishes have emerged as the single most important factor causing the decline of Colorado River basin fishes (e.g., Mueller and Marsh 2002, Clarkson et al. 2005, Marsh and Pacey 2005, Propst et al. 2020).

The Colorado River (Fig. 1) is nearly 1450 mi (2334 km) long and drains about 246,000 square miles (637,137 km²) of the western United States and northwestern Mexico. The Lower Colorado River is the focus of this paper. It is defined for political purposes as the mainstem reach downstream from Glen Canyon Dam (Lake Powell) to the river delta in Mexico. The Grand Canyon region that divides the basin (i.e., the Colorado River watershed, which includes tributaries such as the Gila River) also is geologically significant because the upper and lower basins became integrated here. As crustal extension and mountain building diverted the river south toward the Sea of Cortez (Gulf of California), the Grand Canyon began to down-cut as the Colorado Plateau uplifted, and the 2 basins connected ca. 5 Ma (Minckley et al. 1986, Spencer et al. 2008). The Colorado River watershed is mostly arid and includes the lower basin in Arizona, California, Nevada, and Sonora and Baja California

Norte, Mexico, and the upper basin in Colorado, New Mexico, Utah, and Wyoming. Major tributaries of the lower basin are Paria, Little Colorado, Virgin, Bill Williams, and Gila rivers.

Habitat alterations beginning early in the 20th century have been dramatic. The mainstem was parsed by multiple agricultural and multi-use, low-head diversion structures and massive, high, hydroelectric dams and their artificial reservoirs (see United States Bureau of Reclamation [Reclamation] 1980 for a complete list of structures, capacities, and completion dates). Hypolimnetic releases from larger impoundments are perennially cold, and clear, massive lakes behind dams represent novel habitats historically unknown to the region (Minckley et al. 1986). The hydrograph is highly modified, and flows are regulated in response to electrical, agricultural, and urban demands. Much of the channel downstream of Blythe, California, has been dredged, channelized, and constrained by riprap levees, and once-abundant backwater and other off-channel connectives (hereafter backwater or off-channel habitat [OCH]) have largely been eliminated (Minckley 1979, Brown 1983, Mueller and Marsh 2002). The native ichthyofauna associated with the Lower Colorado River comprised 12 warm-water species (Table 1), 2 of which were primarily marine (Minckley 1979, 1991). While human modifications altered the physical, chemical, and hydrological character of the river, the introduction and establishment of dozens of nonnative fish species polluted the system (Dill 1944, Miller 1961). Half of the native species in the Lower Colorado River are federally listed as endangered and 5 are extirpated. The community is now dominated by nonnatives in forever-altered habitats.

A BRIEF HISTORY OF FISHERIES MANAGEMENT ON THE LOWER COLORADO RIVER

Lower Colorado River fishery management has focused on recreation since the first stockings of sport species in the 1930s (U.S. Fish and Wildlife Service [USFWS] 1981). The 5 resident “large river” native species—bonytail *Gila elegans* (Photo 1), humpback chub *Gila cypha*, Colorado pikeminnow *Ptychocheilus lucius*, flannelmouth sucker *Catostomus latipinnis*, and razorback sucker *Xyrauchen texanus* (Photo 2)—were ignored. Although impacts of water development and introduced fishes on the large river fishes were recognized relatively early

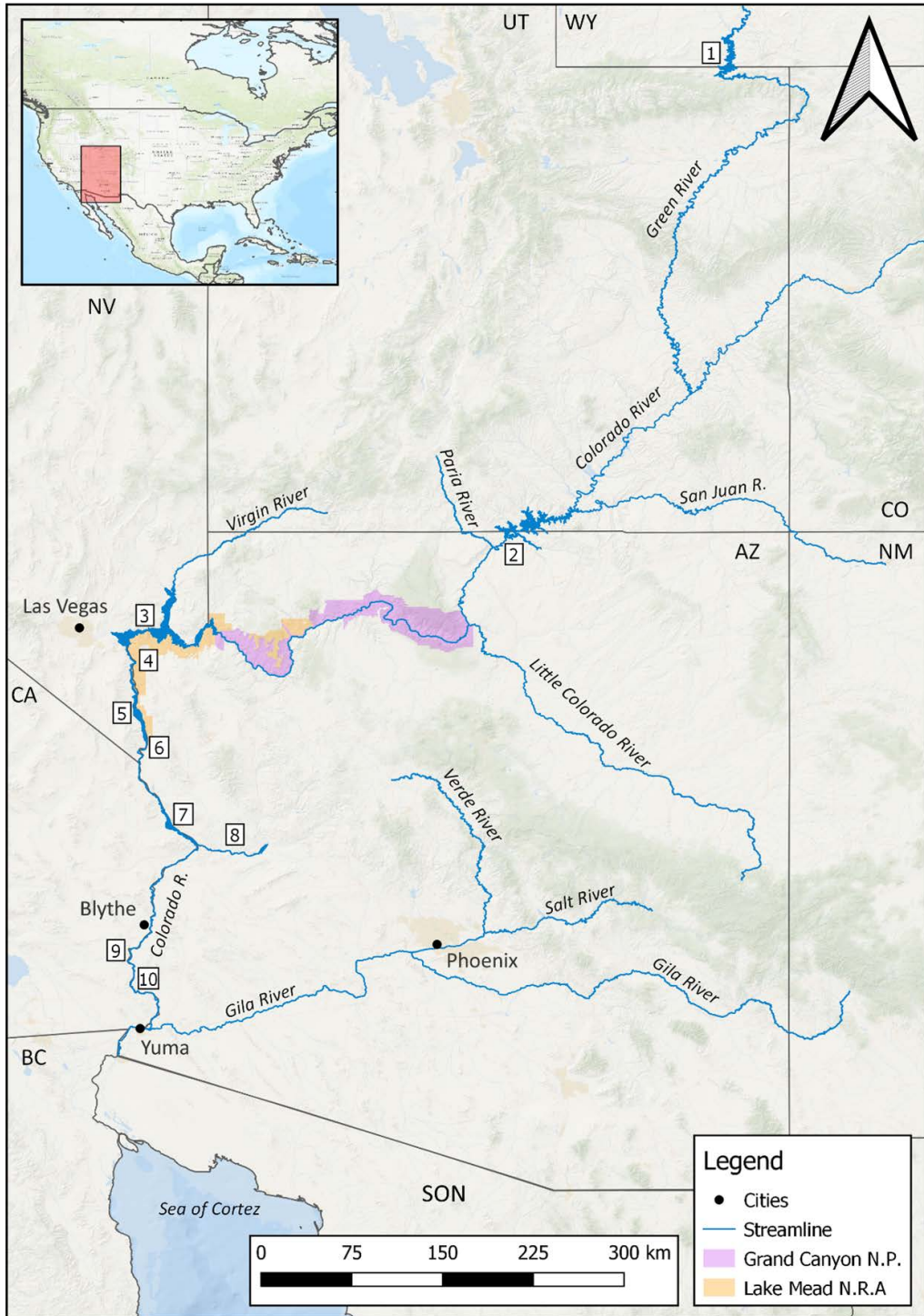


Fig. 1. Colorado River basin map showing place names mentioned in the text, and location map (inset). Numbers represent the following, north to south: 1, Flaming Gorge Reservoir and dam; 2, Lake Powell (Glen Canyon Dam); 3, Lake Mead (Hoover Dam); 4, Willow Beach National Fish Hatchery; 5, Lake Mohave (Yuma Cove and AJ backwaters, AZ; Dandy, Nevada Larvae, North Chemehuevi, North Nine-Mile, and Willow backwaters, NV); 6, Davis Dam; 7, Bill Williams River; 8, Lake Havasu (Parker Dam); 9, Cibola High Levee Pond (Cibola NWR); 10, Imperial Ponds (Imperial NWR).

TABLE 1. Common and scientific names of native fishes of the mainstem Lower Colorado River with their current conservation status, listing status under the Endangered Species Act, and comments on distribution.

Common name	Scientific name	Status	Listing status	Comments
Pacific tenpounder	<i>Elops affinis</i>	Locally common	Not listed	Marine; lowermost reach of the river
Humpback chub	<i>Gila cypha</i>	Locally common	Endangered	Grand Canyon only
Bonytail	<i>Gila elegans</i>	Extirpated	Endangered	Functionally extinct in the wild, supported only by stocking
Roundtail chub	<i>Gila robusta</i>	Extirpated	Not listed	Restricted to tributaries
Woundfin	<i>Plagopterus argentissimus</i>	Extirpated	Endangered	Restricted to Virgin River
Colorado pikeminnow	<i>Ptychocheilus lucius</i>	Extirpated	Endangered	Upper Colorado River basin only
Speckled dace	<i>Rhinichthys osculus</i>	Locally common to abundant	Not listed	Grand Canyon and tributaries only
Flannemouth sucker	<i>Catostomus latipinnis</i>	Locally common to abundant	Not listed	Grand Canyon only; repatriated below Davis Dam
Bluehead sucker	<i>Pantostomus discobolus</i>	Locally uncommon to common	Not listed	Grand Canyon and tributaries thereto
Razorback sucker	<i>Xyrauchen texanus</i>	Rare to locally common	Endangered	Supported only by stocking ^a
Striped mullet	<i>Mugil cephalus</i>	Common	Not listed	Marine; lowermost reach of the river
Desert pupfish	<i>Cyprinodon macularius</i>	Extirpated	Endangered	Restricted in the wild to remnant populations in California, Baja, and Sonora.

^aA razorback sucker population of uncertain origin in Lake Mead and western Grand Canyon includes fish reported to have recruited in situ (Albrecht et al. 2010).

(Dill 1944, Miller 1946, Jonez and Sumner 1954), there was little interest in conserving native species. Absence of conservation management exacerbated these impacts. Thus, Colorado pikeminnow was extirpated from the Lower Colorado River, and the entire lower basin (the last record was in 1967 from Lake Mead) and flannemouth sucker downstream from Grand Canyon was not seen after 1973 (Minckley and Marsh 2009, unpublished data). Bonytail was functionally extinct range-wide by the 1980s (Kaeding et al. 1986, Marsh 2004), and razorback sucker in the lower basin was reduced to a single, nonrecruiting population in Lake Mohave (Minckley 1983, Marsh and Minckley 1989). An important “wild” population of a few hundred razorback suckers of uncertain origin persists in Lake Mead (Albrecht et al. 2010). This population was thought to have been extirpated by the 1970s but has shown evidence of natural recruitment, otherwise now essentially unknown for the species, and positive population growth in the initial years after its reappearance in the early 1990s (Albrecht et al. 2010). Of the “large river” fishes, only humpback chub has fared relatively well in altered habitat, but until recently the species was restricted largely to Little Colorado and its confluence with Colorado River in Grand Canyon (Kaeding and Zimmerman 1983, Valdez and Ryel 1995). Recent shrinkage of Lake Powell and increased water temperature downstream have allowed humpback chub to expand its range into western Grand Canyon where it now thrives (Van Haverbeke et al. 2017, Rogowski et al. 2018). The overall deteriorating status of these iconic fishes was documented throughout the Colorado River basin (Miller 1961, Minckley and Deacon 1968), and humpback chub and Colorado pikeminnow were on the first list of endangered species (USDI 1967), to be joined by the equally imperiled bonytail in 1980 (USFWS 1980) and, after a decade-long politically inspired delay, razorback sucker in 1991 (USFWS 1991a).

Why Bonytail and Razorback Sucker?

We restrict further coverage here to bonytail and razorback sucker because of their persistence and precarious state in the Lower Colorado River. Bonytails are rare and thus little studied, so there are few contributions to a review of the species’ biology (Pacey and Marsh 1998). Razorback suckers, on the other hand, have been studied extensively, and their



Photo 1. Bonytail, *Gila elegans*. B. Gratwicke, Smithsonian Conservation Biology Institute.

life history and ecology are treated in several excellent reviews (Minckley et al. 1991, Marsh et al. 2015, Bestgen et al. 2020) and in the species' SSA (= species status assessment; USFWS 2018). Dramatic reduction in distribution and population sizes of both species led the Lower Colorado River Multi-Species Conservation Program (LCR MSCP¹) to prioritize these fishes for conservation management in the Lower Colorado River mainstem, and actions on their behalf have been actively and aggressively implemented. We do not deal with Colorado pikeminnow (formerly known as Colorado squawfish) because it was absent from the Lower Colorado River when the program was implemented and remains so. Furthermore, there were objections to stocking Colorado pikeminnow that would be fully protected in the Lower Colorado River under the Endangered Species Act (ESA); these objections centered largely around potential influence on river operations and existing sport fisheries of a large, endangered, predatory species. As a result, "experimental non-essential" status for Colorado pikeminnow was rejected despite that designation for fish stocked into the Salt and Verde

rivers in central Arizona (USFWS 1985, Hendrickson 1993). Humpback chub also is not a covered species because it historically and currently occurs only in Grand Canyon upstream of the LCR MSCP geographical area. Finally, flannelmouth sucker, while an LCR MSCP-covered species monitored by the program, is not a species of concern and not actively managed. Bonytail, humpback chub, Colorado pikeminnow, and razorback sucker all are actively managed in the Upper Colorado River basin under 2 recovery implementation programs covering that portion of the species' range (USFWS 1987a, 1991b).

Applied Fisheries Management

Traditional fish management tools that could be used to benefit bonytail and razorback sucker include population control, stocking to augment existing populations or create new ones, habitat manipulation to improve existing conditions or create new places for the fish to live, regulation, and public education, with the last 2 actions targeted at human uses and users. We summarize below how these tools have been applied (or not) in the Lower Colorado River.

¹Critical habitat for bonytail and razorback sucker was designated in 1994. Soon thereafter, the agencies that manage water and power on the Lower Colorado River met to discuss plans to conserve these and other native species and their habitats in compliance with the ESA. A 1997 USFWS biological opinion called for stakeholders to develop and implement the LCR MSCP, and in 2005 the implementing documents were signed by the Department of the Interior and the states of Arizona, California, and Nevada (see www.lcrmscp.gov). The lead agency is the U.S. Bureau of Reclamation, and the LCR MSCP covers 27 species including 4 fishes plus other aquatic and terrestrial biota. Guidance is provided in a habitat conservation plan (LCR MSCP 2004) and annual work plans.



Photo 2. Razorback sucker, *Xyrauchen texanus*. © J. Sartore/Photo Ark.

Selective harvest, removal, or chemical treatment to control nonnative predators to reduce impacts on native species could benefit native fishes, but it has been applied only locally in select, isolated habitats in the Lower Colorado River basin. In the Upper Colorado River basin, the exact opposite was practiced. The Green River poisoning was implemented in 1962 to eliminate native fishes, primarily minnows (Cyprinidae) and suckers (Catostomidae) in advance of an anticipated sport fishery for trout in the newly created Flaming Gorge Reservoir (Holden 1991). Since then, mechanical removal of predatory nonnative fishes is actively practiced in Grand Canyon and the Upper Colorado River basin to benefit native species. These 2 programs, the Green River poisoning and current predator control, half a century apart, serve as an example of the gradual change that began in the 1980s in management agency mindset, at least superficially, from “sport fish at all costs” to “native fish despite the cost.”

Bonytail and razorback sucker were brought early into the federal hatchery system where efficient methods of artificial propagation and production were devised (e.g., Toney 1974, Hamman 1982, 1985). Repatriation was the management emphasis for both species—a necessary action because naturally recruiting wild populations were unknown. Bonytail repatriation has occurred primarily in the Colorado River and adjacent habitats. Other than dead-end

stocking into a few urban ponds and lakes, no effort has been directed toward its expansive historical habitat in the Gila River basin. Between 1991 and the end of 2022, bonytail stocking totaled 483,224 fish: 98,172 into Lake Mohave and lakeside backwaters (the mouths of inundated desert washes separated from the reservoir by wind-generated gravel berms), 284,392 into Lake Havasu and adjacent habitats, 99,591 into the Lower Colorado River mainstem and its connectives downstream of Lake Havasu, and 1069 into inland impoundments and isolated backwaters with no river access. Bonytail repatriations into open waters have produced disappointing results (i.e., no evidence of long-term survival, reproduction, or recruitment; Humphrey et al. 2016, Heishman et al. 2022), while stockings into habitats free of nonnative fishes have been encouraging because survival, reproduction, and recruitment occur. Studies of life history and ecology were facilitated by long-term survival in Cibola High Levee Pond (Photo 3, Appendix 1.1) in Arizona and California (Mueller 2006) and in constructed native fish ponds at Imperial National Wildlife Refuge (NWR) (Imperial Ponds hereafter; Photo 4 and Appendix 1.2) near Yuma, Arizona (e.g., Kesner et al. 2008). Management at both sites comprised provision of spawning substrate, repeated stockings, and various attempts to control nonnative fishes. Work at Imperial Ponds included attempts to maintain suitable habitat conditions,



Photo 3. Cibola High Levee Pond, La Paz Co., Arizona, and Riverside Co., California. An example of a seminatural, off-channel habitat on a river channel. C. Ehlo, USFWS, November 2016.



Photo 4. Imperial Ponds, Yuma Co., Arizona. An example of a wholly artificial, off-channel pond complex. Ponds are numbered sequentially 1 to 6 from top to bottom in this photo. U.S. Bureau of Reclamation, October 2009.

specifically adequate summer oxygen. Successful life-cycle completion and establishment of multiple year classes were documented at both sites. Ironically, bonytail is easily propagated artificially and reproduces prolifically in captivity and in sites absent nonnative fishes.

However, attempts to establish new populations through stocking into open waters or closed systems where nonnative fishes are present have consistently failed due to high poststocking mortality from avian and piscine predation that present a suite of mitigation challenges

(e.g., Carpenter and Mueller 2008). Establishing wild populations of bonytail and finding alternatives to hatchery production and repatriation to perpetuate this critically imperiled species are among the most urgent native fish conservation challenges in western North America.

Razorback sucker management in the Lower Colorado River is summarized in a series of papers published over 4 decades (Minckley 1983, Minckley et al. 1991, Marsh et al. 2015, Bestgen et al. 2020). Most activities paralleled those directed toward bonytail, and the 2 species often were co-treated in the same habitats. Razorback sucker was initially brought into captivity in 1974 when a sample of wild fish from Lake Mohave was taken to the federal hatchery at Willow Beach, Arizona, where it was subjected to propagation studies. Initial successes led to a decision to transport, in 1981 and 1982, the first brood stock to the USFWS native fish facility at Dexter, New Mexico, now named Southwestern Native Aquatic Resources and Recovery Center (SNARRC). Hatchery production and rearing were ramped up quickly to support an ambitious program of repatriation to Gila River basin streams facilitated by a memorandum of understanding between Arizona Game and Fish Department and USFWS that provided for 10 years of stocking into central Arizona waters in lieu of listing under the ESA. Despite stocking millions of young fish, the program failed to establish any new populations because long-term survival was absent and no reproduction occurred (Marsh and Brooks 1989, Minckley et al. 1991, Schooley and Marsh 2007). Consequently, the species was listed as endangered in 1991 (USFWS 1991a). At the same time, thousands to tens of thousands of hatchery-produced fish were distributed to Arizona, California, and Nevada, to aquariums and zoos, and to federal facilities to support broodstock development, propagation and rearing, research, and stocking into captive and “natural” sites. Initial repatriations to the Lower Colorado River channel were in 1980, followed by stockings into lakes Havasu (1986), Mohave (1992), and Mead (1995). Between 1991 and the end of 2022, 725,837 razorback suckers had been stocked: 296,875 into Lake Mohave and lakeside backwaters, 191,370 into Lake Havasu and adjunct habitats, 228,154 into the Lower Colorado River mainstem and its connectives downstream of Lake Havasu, and 9438 into inland impoundments and isolated backwaters with no river access.

Native fish management goals under auspices of the 50-year LCR MSCP initiated in 2005 include stocking 620,000 bonytails and 660,000 razorback suckers into the Lower Colorado River, maintenance of a razorback sucker broodstock and genetic refuge (in Lake Mohave), and creation or protection of 360 acres (146 ha) of habitat for bonytail, flannelmouth sucker, and razorback sucker (LCR MSCP 2004). Through federal fiscal year 2022, 131,987 bonytails and 273,676 razorback suckers had been stocked and 158 acres (64 ha) of habitat, referred to as “conservation areas,” had been created or protected (James Stolberg, Bureau of Reclamation, personal communication, January 2023). Stocking implementation is reviewed, evaluated, and updated periodically (e.g., LCR MSCP 2020).

Habitat manipulation to benefit bonytail and razorback sucker has largely been restricted to off-channel connectives and isolated, adjacent waters; no such actions have been implemented in the river channel proper or its reservoirs. Reasons for this include a lack of practical strategies to carry out such activities and limited support from the states, which preferentially support recreational sport rather than native nongame fisheries; the two activities are not compatible (Clarkson et al. 2005). Ephemeral and perennial isolated and open coves along the shorelines of lakes Mohave (e.g., Appendix 2) and Havasu have been modified by deepening, berm construction, netting emplacement, and installation of wind turbines to help maintain suitable temperature and dissolved oxygen (DO). These sites support seasonal grow-out and life-cycle completion (reproduction and recruitment) and represent sources of fish for repatriation and opportunities for study; several have been used for research on bonytail and razorback population genetics and demography (Appendices 1, 2).

Other manmade sites (e.g., golf course ponds, urban lakes, and stock waters) also have been used for holding or grow-out, and some are also important study areas. For example, Cibola High Levee Pond (Appendix 1.1), an isolated segment that was cut off when the river was channelized and riprapped in the latter 1960s, represents an important site of observation, experimentation, and discovery (e.g., Mueller et al. 2005). In addition, native fish ponds at Imperial Ponds (Appendix 1.2), initially created for waterfowl and later reconfigured into 6 small lakes, can support razorback sucker and bonytail in either mono- or biculture situations where

drivers of population dynamics such as survival/mortality, recruitment, growth, etc., can be investigated. Opportunities to examine effects on native fishes of vegetation removal or enhancement, provision of cover, control of avian or other predators, or other factors are available and awaiting application both at Cibola and Imperial Ponds.

Harvest of bonytail and razorback sucker by any means is prohibited under the ESA and state rule, except for scientific purpose and incidental take. Thus, imposition of other, restrictive regulations such as quotas, seasons, sizes, etc., are unnecessary management tools. Neither species is known for its desirability as a recreational species, and neither is especially susceptible to angling, although there are infrequent reports of fishermen bringing in a carcass, usually hooked in the case of bonytail or snagged in the case of razorback sucker, for examination. Angler reports can be an important source of information, and these fish are generally known to those who fish along the Lower Colorado River and its reservoirs, but we suspect most are unwilling to bring in angled fish because they wish to avoid misperceived consequences of harming an endangered species.

Public education is a powerful management tool (e.g., Fien et al. 2001) that can be used to aid in the conservation of bonytail and razorback sucker. Each of the basin states prominently features bonytail, razorback sucker, and other native species on its website, in its license-related regulations, and as part of promotional literature and “trinkets.” Information and education (I&E) are integral aspects of state and federal agency services, and programs like the LCR MSCP have dedicated if not well-funded I&E. A public school program for kindergarten through third grade, “Sharing Tails,” combines a live, in-person or virtual presentation with crafts and learning games to foster knowledge of native fishes and an interest in their conservation. These outreach efforts have been successful in making a segment of the general public aware of bonytail and razorback suckers, but unfortunately, these fishes are still regarded by many as expendable. Many anglers are interested in their preferred sportfish and seem to lack an appreciation and acceptance of species like bonytail and razorback sucker for their unique place and role in the natural world. Resources of time and money directed toward these fish may be perceived by some as wasted

when those same resources could have been applied toward management of sport species (Clarkson et al. 2005). Public education toward development of a positive conservation ethic represents our ultimate challenge, and one that must be met to preserve species like Lower Colorado River native fishes and the biodiversity to which they contribute.

GENESIS OF AN OFF-CHANNEL HABITAT CONCEPT

Concerns over the deteriorating status of bonytail and razorback sucker in the Lower Colorado River basin led to the development of a plan focused on reversing population declines, stabilizing existing populations, and eventually establishing and maintaining self-perpetuating populations in this part of the basin (USFWS 2005). Six objectives were identified and discussed, starting with prevention of extinction (level I) and culminating in recovery (level VI); these were summarized in a flowchart (Fig. 2) designed to direct management actions for conservation of these fishes (Minckley et al. 2003, see also USFWS 2005).

The OCH management concept was developed in the late 1990s. At the time, bonytail was gone and traditional approaches had failed to stem the decline of razorback sucker populations throughout the Lower Colorado River (above, reviewed in Marsh et al. 2015). Concerns over the status of razorback sucker have prompted informal meetings of scientists and managers since the 1970s, and these evolved into formalization in 1990 of the Lake Mohave Native Fish Work Group (hereafter NFWG, Mueller 1995), which focused its attention on fish in that reservoir. Two major observations shaped the approach of the NFWG: (1) the primary cause of failed recruitment was predation by nonnative fishes on larvae (Minckley 1983, Minckley et al. 1991, Mueller 1995) and (2) maintenance of genetic diversity was important for long term persistence (reviewed in Minckley et al. 2003). Nutrient limitation also was carefully examined for its potential impact on recruitment (Papoulias and Minckley 1990, 1992, Horn 1996), but its effects were deemed insignificant compared with predation. The NFWG objective was to replace a wild population of about 50,000 adult fish in Lake Mohave that existed in the late 1980s with repatriates that represented genetic diversity found in the original wild population. Razorback sucker individuals can live for 40

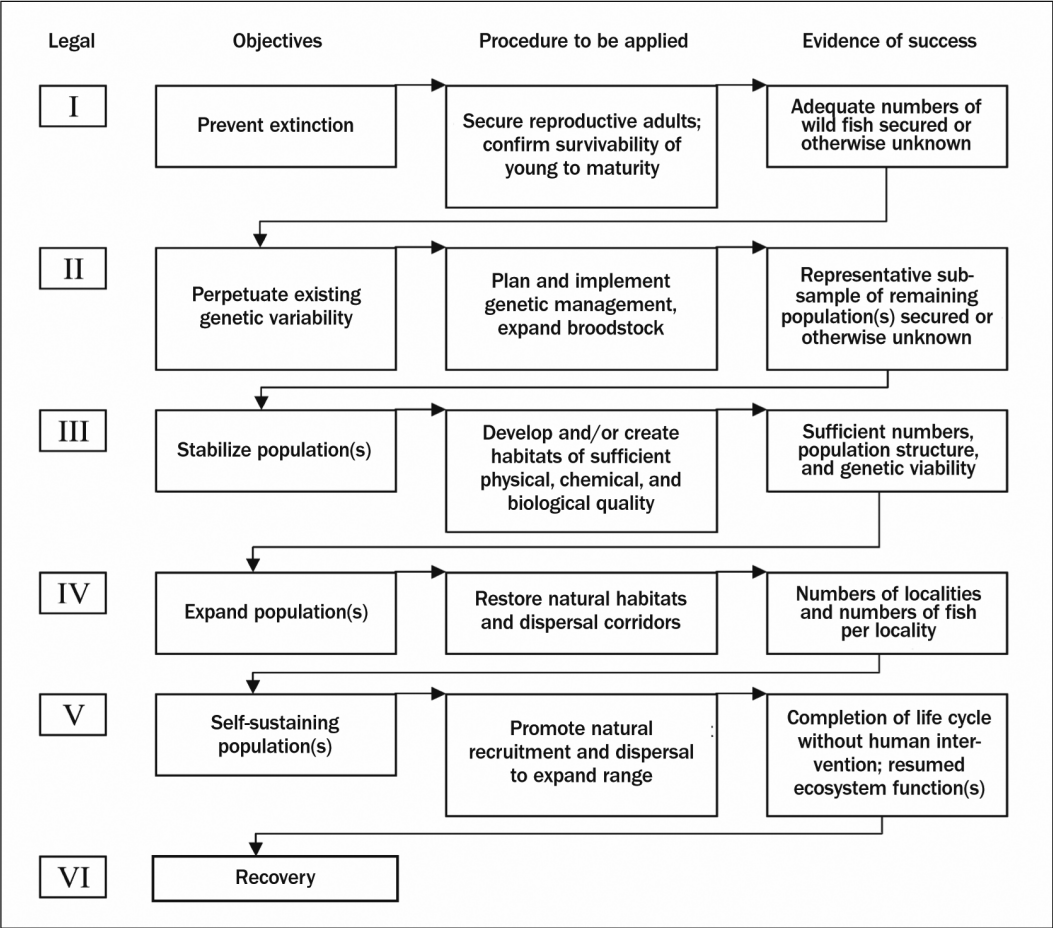


Fig. 2. Conceptual plan for managing native fishes of the Lower Colorado River basin (figure 3 from Minckley et al. 2003).

years or more, and a single female can produce >200,000 ova in a season; hence, this population could provide more time for managers to solve the recruitment issue in the open river mainstem.

Because predation by nonnative fishes, especially on larvae, was the primary cause of recruitment failure in both species, this life history stage and impacts thereon were bypassed by stocking larger individuals that could survive to become adults. This stocking could have been accomplished through hatchery spawning, rearing, and augmentation; however, concerns about the effects of hatchery production on genetic diversity and inbreeding (e.g., Dowling et al. 1996) led to development of alternative methods for replacing the wild razorback sucker population in Lake Mohave. Production of genetically diverse progeny initially involved the use of

Yuma Cove backwater, an isolated lakeside OCH (e.g., Appendices 1.3, 2) that was free of nonnative fishes. Various approaches to seeding this backwater were explored and evaluated including natural spawning of adults and stocking of fertilized ova and larvae (reviewed in Marsh et al. 2015). The program ultimately selected, and still uses, a strategy of collecting larvae that are naturally produced in the lake and then transferring them for rearing into protected facilities such as hatchery raceways, backwater ponds, or other protected off-site locations. Subadult fish are stocked back into the lake after growth to sizes thought to be large enough to evade predators (Mueller 1995).

Hatcheries, whereby individuals reside in captivity from birth until their release as adults, and supplemental stockings are the cornerstones of many conservation programs, despite

their well-documented negative effects (e.g., Frankham 2008, Stringwell et al. 2014, Fraser et al. 2018, Horreo et al. 2018, Blouin et al. 2021) and limited evidence of recovery of hatchery-supported populations (e.g., Thiem et al. 2017). Importantly, the NFWG members were cognizant of avoiding the shortcomings and negative effects of hatchery propagation for the Colorado River's large river fishes. Nowhere in open waters of the Lower Colorado River do hatchery-reared bonytails persist other than briefly after stocking (Pacey and Marsh 2008, Humphrey et al. 2016, Kesner et al. 2017, Kelley et al. 2021) nor do, other than in isolated, protected habitats, the vast majority of razorback suckers (Schooley and Marsh 2007, Karam et al. 2008, Marsh et al. 2015). As such, it is not feasible (nor possible in the case of the bonytail) to generate large populations of these fishes in riverine or reservoir habitats populated by predatory nonnative fishes.

These issues led to development and formalization in the late 1990s of the OCH concept presented in Minckley et al. (2003). They concluded that it would not be possible to recover these species under available conditions, or to even develop self-sustaining and expanding populations in the historical range. Despite this pessimism, they believed it was possible to stabilize populations (level III, Fig. 2) with continued assistance from managers. It is notable that stabilization was not defined simply in terms of specific population sizes but also considered "the genetic variability that existed a century ago in native big-river fish populations." Population goals were discussed in terms of number of breeding adults necessary for a stable population, with estimations made using comparisons with similar species and knowledge of genetic and demographic parameters.

The OCH concept proposed establishment of self-sustaining populations in multiple predator-free habitats, with individuals from these waters subsequently used to maintain populations in the mainstream Lower Colorado River. Genetic diversity would be maintained in both riverine and off-channel populations through periodic, two-way exchange of individuals, increasing the effective number of breeders at each location and preventing divergence due to stochastic processes. Discussion of the plan included potential design alternatives for OCHs, population goals, and necessary management actions. Unfortunately, little was known about the number

of individuals and number of breeders that could be successfully maintained long-term in backwater and riverine situations, or the dynamics of such populations; however, examples from Cibola High Levee Pond and the Yuma Cove backwater encompassed a range of situations that supported the plausibility of the approach.

Implementation of the OCH concept is still in its development and experimentation phase with focus on understanding the demographic and genetic dynamics of fish in these small systems. The ideas in Minckley et al. (2003) were formalized as part of recovery plans for large river fishes (USFWS 2005), yet the OCH program has been slow to develop, due partly to difficulties securing habitat. Challenges to full implementation stem, to a certain extent, from the uniqueness of an approach not readily accepted by all stakeholders, reflecting both a failure to recognize the importance of putting these principles into practice for long-term management success and the allure of traditional (i.e., hatchery-based) approaches. In addition, LCR MSCP compliance requires creation or development of a relatively small total surface acreage of OCH, and there is no requirement for these habitats to be for natives only, nor is there a requirement for native fish to persist in these places. This is especially problematic because state agencies do not readily relinquish any location where they see opportunity for sport fishing.

MATURATION OF THE OFF-CHANNEL HABITAT CONCEPT—FILLING CONCEPTUAL AND RESEARCH GAPS

The OCH plan (Minckley et al. 2003) motivated an intensive research program aimed at addressing uncertainties that impeded its implementation, including exploration of the importance of environmental, demographic, and genetic information. This information is critical for design and successful implementation of OCHs that hold self-sustaining populations.

Minckley et al. (2003) suggested that the size for each OCH should be between 2 and 5 surface acres (about 1 to 2 ha). There were scant data on habitat size requirements for either species at the time, but this size was considered large enough to sustain a viable population of genetically diverse fish yet small enough to manage (drain, renovate, and sample/harvest). Larger ponds may allow for larger self-sustaining populations, but they are more difficult to monitor,



Photo 5. Yuma Cove backwater, Mohave Co., Arizona. An example of a natural, off-channel habitat on a reservoir. U.S. Bureau of Reclamation, February 2008.

maintain, and mitigate against contamination by nonnative fishes (e.g., Kesner et al. 2012). A quantitative relationship between habitat size and population size has yet to be determined, and there may be an unavoidable conflict if adult population sizes must be larger than a few hundred individuals due to demographic or genetic concerns, or if populations become resource limited shortly after stocking (e.g., Yuma Cove backwater—Photo 5 and Appendix 1.3).

The LCR MSCP has used more than a dozen waters along the Colorado River for grow-out of razorback sucker, and others were repurposed for use in the OCH program for both species. Yuma Cove backwater (Appendix 1.3) was used for grow-out of razorback sucker and has contained an experimental population of this species since 2013. Larger ponds (>1 to 21 acres [>0.4 to 8.5 ha]) predominate elsewhere along the system to meet LCR MSCP habitat restoration/creation goals (LCR MSCP 2004), and those at Imperial Ponds also have been converted for the OCH program. These ponds originally were developed in 2000 for waterfowl and native fish and were set aside for razorback sucker grow-out from 2002 to 2005, and then redeveloped as 6 independent ponds. These OCH ponds have been extensively monitored for water physicochemistry, benthic and planktonic

fauna, and fish survival, growth, and recruitment (Kesner et al. 2012, Dowling et al. 2021). Results of these studies have furthered our understanding of how OCH size and design features impact the ability to sustain conditions favorable to bonytail and razorback sucker; the ponds now contain 3 populations of each species.

Regardless of habitat size, environmental conditions must be maintained within tolerable limits to sustain populations of bonytail and razorback sucker. As originally envisioned by Minckley et al. (2003), water levels in OCH ponds would be maintained by passive flow through a porous berm; the bottom elevation of the backwater would be below the elevation of the adjacent river to ensure constant water supply. Water levels in Cibola High Levee Pond and Yuma Cove backwater are maintained effectively through such a system, but backwaters specifically built for the LCR MSCP generally have not relied on passive flow alone, but have actively pumped some water, because the passive flow system could not always support adequate physicochemical conditions year-round (USBR 2005, LCR MSCP 2008). Razorback suckers raised in Imperial Ponds from 2002 to 2005 experienced die-offs in summer months (USBR 2005). The die-offs, resulting from high temperature and low dissolved oxygen

(DO) in summertime, were partially ameliorated by pumping well water into the ponds. The pumping of well water did not prevent another die-off in 2004. It was decided that the ponds should be redeveloped with features to improve water physicochemistry and provide habitat for self-sustaining populations of both species (USBR 2005).

During the redevelopment of Imperial Ponds and prior to additional stocking of native species, thresholds for water physicochemical parameters and mitigation strategies (additional water inflow) were proposed based on laboratory experiments (Bulkley and Pimentel 1983, Marsh 1985, Stolberg 2012) and field studies (Minckley et al. 1991, Mueller 2006). When conditions fell outside these thresholds—pH exceeding 9.0, water temperature exceeding 33.3° C, or DO dropping below 4.0 mg/L (Kesner et al. 2008, LCR MSCP 2013)—additional water would be supplied as a mitigation strategy. Little was known prior to the redevelopment about the amount of water required for mitigation other than that the single on-site well appeared to be inadequate as a source. A pump was built and plumbed to provide river water filtered through a wedge wire screen. The wedge wire screen was intended to ensure that viable fish eggs and larvae would not be transferred into the ponds (LCR MSCP 2008). Redevelopment was completed in 2006, and all 6 ponds were stocked with native fish in 2007: 3 ponds with razorback sucker and 3 ponds with bonytail. The pump filter failed to exclude nonnative eggs and larvae (McDonald and Karchesky 2010), and thus, although suitable physicochemical conditions may have been maintained, all 6 ponds included populations of nonnative fishes detrimental to recruitment success of the native species (Kesner et al. 2012).

The initial Imperial Ponds experiment from 2007 through 2012 was not a complete failure. Both razorback sucker and bonytail reproduced, and recruitment of at least one year class was confirmed prior to the establishment of nonnative fishes (Kesner et al. 2012). By the end of 2012, it became apparent that no additional recruitment would occur in the presence of nonnative fish species. After extensive salvage operations to capture and release the remaining native fishes in the ponds, a series of chemical renovations was attempted in all 6 ponds to eliminate nonnative fishes (LCR MSCP 2014). While these efforts generally were successful, multiple treatments were required and western mosquitofish

Gambusia affinis maintained populations that could not be eradicated. This species is known to prey on larval razorback sucker (Ley et al. 2014), but its overall impact on recruitment is still unknown. Imperial Ponds, like other OCHs adjacent to the Colorado River system, were purposely built so that their bottom elevation was below surface water in the river, thus providing a subsurface supply of water. Unfortunately, this arrangement also eliminated the option to remove nonnative fishes (e.g., renovate) by desiccation because such sites could not be drained thoroughly. The subsurface supply also decreased the effectiveness of chemical treatments, as the influx of freshwater provided a chemical refuge. This problem was addressed at Imperial Ponds by pumping water into the ponds prior to renovation to increase pond water levels above the river stage, thereby reversing the subsurface flow, but also increasing the amount of chemical required (LCR MSCP 2014).

During the period of chemical treatments at Imperial Ponds, and subsequent water-level and physicochemical monitoring, additional wells were established to independently supply each pond, and the river pump was disconnected from the system. Imperial Ponds was again stocked with bonytail and razorback sucker in 2017 (LCR MSCP 2020), and recruitment of both species was subsequently documented (unpublished data). Bonytail die-offs occurred in 2020 and 2022, and razorback sucker populations crashed in 2022 (Jeff Lantow personal communication, unpublished data). These occurrences were due to human error or well pump failure and were not necessarily reflective of the effectiveness of established thresholds or mitigation strategy.

As initially envisioned by the design team, Imperial Ponds was meant to meet and test the OCH concept of Minckley et al. (2003) as well as provide additional habitat features hypothesized to be important for either species based on field observations of remaining mainstem populations and other OCH examples (USBR 2005). At Cibola High Levee Pond, adult bonytail were tracked repeatedly to specific crevices along a riprap-lined shore (Marsh et al. 2013). The Imperial Ponds design included large boulder shorelines to provide these crevices. The implementation proved inadequate because a thin veneer of smaller rock failed to provide the interstitial sizes sought by the species and the available crevices filled with sediment (Kesner et al. 2012).

Razorback suckers are known to prefer clean substrate in the size range of large sand to small cobble (Mueller 1989, Minckley et al. 1991). The Imperial Ponds design called for gravel-cobble shorelines to provide spawning substrate throughout the ponds. Implementation of this feature was reduced to gravel boat ramps, and gravel-lined small islands called “hummocks.” After the 2007 stockings and marginal reproductive success, purpose-built spawning beds of cobble were added to 2 ponds with razorback sucker, and recruitment events were observed in both ponds (Kesner et al. 2012). However, the establishment of nonnative fishes and failed recruitment in subsequent years made it difficult to draw conclusions regarding the overall effectiveness.

Limited success at Imperial Ponds provides somewhat of a counterpoint. Failure of the ponds to produce consistent results may be due to their larger size, nonnative fish presence, water mitigation failures, and inadequate habitat features, but we cannot be sure that there is not some environmental condition (e.g., shore gradient) that we are failing to consider.

IMPORTANCE AND RELATIONSHIPS OF GENETICS AND DEMOGRAPHY

When the OCH plan was initially proposed, little was known about genetic diversity (Appendix 3) and its relationship to demography in bonytail and razorback sucker, although maintenance and enhancement of genetic diversity was a foundational goal. Genetic effective size (N_e) is a key metric because it translates observed levels of genetic diversity to numbers of individuals required to maintain that diversity, and thereby directly connects demography (e.g., abundance patterns) to expected levels of genetic diversity. However, the relationship of the effective population size (N_e) and population census size, estimated as the number of reproductively capable adult fish (N_c), depends on the life history (e.g., age at maturity, life span, generation time, fecundity) of individual species. In the plan, legacy genetic diversity for both species was estimated as the average generational time to common ancestry (coalescence) of unique genetic variants that remained in each species. “Coalescence time” is proportional to historical or long-term N_e (Kuhner et al. 1998), and its use yielded long-term estimates of female N_e to be ~90,000 for bonytail and ~700,000 for razorback

sucker (Garrigan et al. 2002). It logically followed that both species must have had large N_e prior to widespread system alteration. Authors of the OCH plan developed guidelines based on theory and expert opinion about the N_e value necessary to maintain legacy genetic diversity for these species, but they were unable to estimate it directly (Minckley et al. 2003).

The ratio N_e/N_c is an important measure of population performance (Palstra and Ruzzante 2008) and is usually <1 in natural populations (Frankham 1995, Vucetich et al. 1997), meaning that more adult fish are required to maintain a target value of N_e . For example, if $N_e/N_c = 0.1$ then $N_c = 1000$ is needed to maintain $N_e = 100$. Based on an educated guess, plan authors proposed that N_e/N_c was ~0.3 for razorback sucker, implying that roughly 3 times the number of reproductively capable adult fish were needed to maintain a target value of N_e . Yet, bonytail and razorback sucker have enormous reproductive capacity per female, and as a result, N_e/N_c can take very low numbers (e.g., $N_e/N_c < 0.00001$ [e.g., Hedgcock 1994]) depending on variance in reproductive success (VRS) among spawners and other factors (Hedrick 2009). Thus, the ratio had to be estimated directly to avoid the possibility of serious underestimation of the N_e value actually needed to meet the goals of the OCH plan.

Given the above issues, management of populations in OCHs requires knowledge of contemporary N_e , not historical or long-term N_e . Contemporary N_e reflects the number of individuals contributing to the gene pool in ecological (i.e., annual or generation), not evolutionary time (Waples 2005). Extensive mark-recapture studies and concurrent genetic sampling of Lake Mohave razorback sucker (Dowling et al. 2014) allowed for direct estimation of contemporary N_e/N_c . Based on changes in mtDNA haplotypes over a 7-year time series (Dowling et al. 2005), the ratio ranged between 0.29 and 0.38 (Turner et al. 2007). Using similar methodology but different genetic markers (microsatellites), Dowling et al. (2014) showed that the N_e/N_c ratio ranged from about 0.1 to 1.0 and increased over a 15-year time series that spanned a transition from wild to repatriated fish, likely because of compensatory effects associated with larval sampling that lowered VRS among adult breeders. Thus, while data were not available for bonytail owing to the scarcity of fish in the wild, these results indicated that the “educated guess”

proposed in the OCH plan was supported by monitoring data for razorback sucker. However, to fully assess the performance of the OCH program, it is necessary to understand factors that affect N_e/N_c in real conditions that capture environmental and demographic fluctuations over ecological time. This was done in a series of studies of ephemeral and permanent ponds.

Ephemeral Pond Studies

Having a robust estimate of N_e/N_c for Lake Mohave razorback sucker was an important step forward, but it was not clear that a similar relationship of demography and genetic diversity in OCHs could be achieved for either species. To address this knowledge gap, a series of experiments was initiated where reproductive adult bonytails and razorback suckers were stocked separately into isolated, ephemeral backwater ponds adjacent to Lake Mohave (Appendix 2) to quantify population genetic and demographic parameters. Prior to stocking, fish were sexed, PIT-tagged, and genotyped using microsatellite markers (Dowling et al. 2014, 2019, Osborne et al. 2018). Roughly 100 male and female parents were stocked into 2 isolated backwaters for each species (Appendix 2), bonytail into Nevada Egg and North Nine-Mile (Osborne et al. 2018) and razorback sucker into Arizona Juvenile (AJ) and Dandy (Dowling et al. 2019).

Harvest data (i.e., fish removed at the end of study) from these short-term (<6-month) experiments were available for razorback sucker stocked in AJ and Dandy (2010–2018) and provided definitive survival information as the OCHs dried during lake drawdowns (Dowling et al. 2019). Harvest was considered complete as ponds were completely dewatered and all fish were removed through netting and seining. Across 9 iterations of the experiment, survival was significantly lower for stocked males than for stocked females. The difference in survival between sexes was partially but not entirely attributed to variation in size at release, with males stocked on average at smaller sizes than females. This sexually dimorphic survival was consistent among years even though survival over the experiment varied temporally, within a backwater among years, and spatially, between backwaters.

Lowest harvest percentages were 6% and 20% for males in Dandy (2016) and AJ (2017) respectively, and 16.7% and 30% for females in Dandy (2018) and AJ (2017), respectively. Highest harvest percentages for males were 68%

(2014) for AJ and 65% (2010) for Dandy, while female harvest peaked at 75% (2014) in AJ and 84% (2010) in Dandy. Mean harvest percentages across all years between the 2 backwaters were similar despite the year-to-year variation, with male harvest percentage at 35% and 31% for AJ and Dandy respectively, and female harvest percentage at 53% and 59% for AJ and Dandy, respectively. Remote sensing of PIT-tagged fish during these experiments provided individualized tracking data for surviving fish in each OCH. This technique was proven effective in contacting 90% to 100% of fish that were later harvested (Dowling et al. 2021). These experiments each occurred over the course of <1 year, so long-term survival and reproductive trends of backwater stocked fish could not be assessed. Results have genetic implications in that skewed sex ratios reduced N_e/N_c and N_b/N_c (N_b = annual effective number of breeders).

Microsatellite markers were characterized for razorback sucker pond samples from 2010 to 2015 and bonytail pond samples from 2014 to 2016. Genotypes from stocked adults and their progeny (Osborne et al. 2018, Dowling et al. 2019) were used to assess maintenance of genetic diversity (e.g., heterozygosity, allelic richness; Appendix 3), determine parentage of recruits, estimate reproductive success and its variance among individual parents (i.e., VRS), and calculate the annual effective number of breeders (N_b), which is a quantity related to N_e (Waples et al. 2011). Progeny were assigned to parents using a statistical algorithm (Jones and Wang 2010), and VRS was calculated for males and females separately (Osborne et al. 2018). For comparison, values are standardized by dividing VRS by mean family size. These data also allowed estimation of VRS for both species.

For bonytail, parentage could be assigned to more than 83%–92% of offspring sampled from North Nine-Mile (2014–2016) and Nevada Egg (2014, 2016) and 73% of individuals sampled from Nevada Egg in 2015. The ratio N_b/N_c was higher for bonytail than for razorback sucker and ranged from 0.72 to 0.86 (Osborne et al. 2018). In 5 of 6 cases, more than 85% of bonytail adults contributed at least 1 offspring to the annual cohort and had correspondingly low VRS. One OCH experienced poor water quality and partial recruitment failure where fewer than 30% of stocked parents contributed to the annual cohort and VRS was relatively high for both sexes (Osborne et al. 2018).

For razorback sucker, the ratio N_b/N_c ranged from 0.09 to 0.68 (unpublished data), which is comparable to N_c/N_c ratios obtained for Lake Mohave (Turner et al. 2007, Dowling et al. 2014). Between 13% and 70% of females contributed at least one offspring to the progeny pool, but reproduction failed in 2 of 5 years in one ephemeral backwater pond. High levels of VRS among ponds and years for razorback sucker are at least partly explained by variation in early mortality of stocked fish. Sex-biased (i.e., male) mortality skews the sex ratio and thereby increases VRS and lowers N_b/N_c . When considered across species, years, and backwaters, experimental results supported the following conclusions: (1) bonytail exhibited higher N_b/N_c and lower VRS than razorback sucker in nearly all cases, (2) razorback suckers performed at least as well in backwater habitats as they did in Lake Mohave in terms of N_b/N_c and VRS, and (3) VRS was attributable to variance in productivity across ponds and years for both species. Rearing in OCHs should preserve legacy diversity at least as well for bonytail as for razorback sucker. However, spatial (e.g., among different backwaters) and temporal (e.g., within backwaters across years) variation in adult survival and reproductive success for both species are still largely unexplained.

Permanent Pond Studies

The short-term experiments in ephemeral pond OCHs followed progeny for 1 year but not recruitment to the adult population. As envisaged by Minckley et al. (2003), the OCH plan calls for age-structured populations that can supply genetically diverse fish to the river mainstem. Questions remain about how well genetic diversity is retained across the life history to recruitment for both species (addressed for Lake Mohave razorback sucker in Carson et al. [2016]). In response, experimental studies have been expanded to include 2 large, permanently filled OCHs: Yuma Cove backwater at Lake Mohave and Imperial Ponds.

Yuma Cove backwater is deep enough to retain water all year, making it an ideal location for establishing a recruiting population. Razorback suckers (100 of each sex) were stocked in 2013 to track multiyear trends in genetic and demographic parameters. Fish were hatchery-reared from wild-caught larvae and at least 2 years old at stocking. Remote PIT-tag sensing was conducted nearly continuously for the first

7 years (February 2013 through March 2019; Dowling et al. 2019). These data were used to track individual survival and incorporated with capture and tagging data to estimate population size using mark-recapture methods. As with the ephemeral pond studies for AJ and Dandy, post-stocking survival was significantly higher for females than for males, even after accounting for differences in release size (Kesner et al. 2019). Razorback suckers stocked in 2013 and known to be alive by February 2014 included 21 of 100 males and 78 of 100 females. Apparent annual survival increased significantly in subsequent years, with 19 males and 75 females surviving through 2017. Additional fish stocked in 2014 and 2015 experienced higher post-stocking mortality, with fewer than 10 from each stocking of 100 fish (50 of each sex in each year) surviving beyond the first year postrelease. This likely was due to the reproductive success of the initial stocking cohort in 2013 and 2014. The loss in overall abundance after the 2013 stocking was replaced by natural recruitment by 2016, and the small OCH maintained an adult population of approximately 400 fish through 2021.

Genetic characterization, parentage assignment of larvae and juveniles, and patterns of reproductive success for razorback sucker were completed for Yuma backwater from 2013 to 2015 using microsatellites as discussed above for ephemeral ponds (Dowling et al. 2019). Parentage assignments worked well for the initial year class of 2013 (97% of parents assigned correctly); however, assignments became more difficult in 2014 and 2015 (77% and 84%, respectively). This likely reflects reproduction by recruits (i.e., fish produced in the backwater), with microsatellite markers providing insufficient statistical power to assign parentage due to shared ancestry of stocked adults and related and uncharacterized recruits.

To further evaluate the longer-term dynamics of survivorship and reproduction in these habitats, the 6 Imperial Ponds were each stocked in 2017 with 150 male and 150 female adult bonytails or razorback suckers (total of 3 ponds for each species—900 individuals per species). After the initial stockings, substantial reproduction at varying levels occurred in all bonytail ponds. Parentage analysis using microsatellite markers revealed high (90%–96%) reproductive success and consequently large genetic effective size ($N_e = 252$, calculated using the sibship assignment

method of Jones and Wang [2010]) for bonytail in one pond—consistent with results from ephemeral ponds. Reproduction was more limited in remaining bonytail ponds, but genetic diversity was preserved between parental and progeny generations, and there was no increase in measures of inbreeding (e.g., F_{IS} , IR; Appendix 3) between adults and progeny. One bonytail pond demonstrated potential issues with using closed habitats for reproduction including limited reproductive contribution of adults, small effective population size, and a subsequent increase in inbreeding measures between parental and progeny generations. Reproduction was limited in razorback sucker ponds for unknown reasons, preventing analyses like those performed for bonytail.

The larger size of ponds and the establishment of age-structured populations at Imperial Ponds pose some additional challenges including sampling sufficient larvae and recruits (i.e., potential parents of subsequent generations) to facilitate parentage analysis and to accurately estimate reproductive success of adults. Alternative measures of genetic “risk” (i.e., increases in inbreeding) could be used to monitor the genetic health of off-channel populations that leverage the power of next-generation sequencing (NGS) methods of genetic monitoring (e.g., genotyping-in-thousands sequencing [GTseq]; Campbell et al. 2015) developed recently for both bonytail and razorback sucker. Baseline values for these measures could be obtained from the time of stocking and thresholds developed that trigger management actions such as movement of fish between habitat complexes. Genomic measures of inbreeding include multilocus heterozygosity (Coltman et al. 1999), internal relatedness (Amos et al. 2001), and relatedness (Wang 2022) (Appendix 3).

REVISION OF THE OFF-CHANNEL HABITAT CONCEPT

In the 20 years since publication of Minckley et al. (2003), a substantial body of research in the areas of demographics, ecology, and genetics has provided important insights pertinent to management of OCHs. This information, in conjunction with studies of bonytail and razorback sucker in OCHs (reviewed above), necessitates revision of the concept as initially envisioned to allow its informed implementation as part of a conservation program for these

species. At the same time, new policies and programs were developed and promulgated, and species conservation goals have been refined and updated. Here we identify specific objectives of an OCH program as they pertain to demographic, ecological, and genetic aspects of OCHs and evaluate specific actions for each species that are dictated by the relatively small size of these habitats.

The OCH concept might be implemented under a conservation program such as the LCR MSCP or a recovery program under the ESA, but there are fundamental differences between the two that are important to contrast. The goal of the LCR MSCP is to develop and implement a plan that will conserve habitat and work toward the recovery of threatened and endangered species, allow existing water removal and power production to continue and optimize future opportunities for these uses, and support incidental take authorizations (i.e., make allowance for accidental or unintentional harm to federally listed species); the goal further includes reducing likelihood that other species might be listed (LCR MSCP 2004). A recovery program is the process by which the conservation status of listed species is improved to the point where they are able to survive on their own in the wild in perpetuity (16 USC § 1531 et seq.). This process includes removal or mitigation of threats in order to halt and reverse the species’ decline and restoration of normal, stable population structure and function such that long-term survival is assured and ESA protection is no longer needed. Species recovery is not only required by a recovery program, but rather is the sole purpose for the program. A conservation program has no such requirement, only that it contributes to recovery. A conservation plan has no expectation of self-perpetuation or removal of covered species from ESA protections. More importantly, the LCR MSCP is in effect for only 50 years, after which—absent renewal or replacement, and successful or otherwise—the conservation requirement ends. Our expectation is that successful implementation of the OCH concept on behalf of bonytail and razorback sucker will contribute to their long-term conservation, but we do not envision recovery for either species.

Ecological and Demographic Considerations— Self-Sustaining and Resilient Populations

A self-sustaining population according to Scott et al. (2005) “should be able to remain stable

or increase over time without human assistance to reproduction or dispersal in the wild.” They add that this condition would be violated by frequent translocations to compensate for human barriers to dispersal or losses due to predation, disease, or other mortality factors but not by occasional translocations to maintain genetic diversity, the latter being an expectation of the OCH approach advocated here.

What a self-sustaining population of bonytail or razorback sucker in an OCH might look like depends on the timescale over which the population is being assessed, that is, whether the assessment is based on a short-term “snapshot” or integrated over many generations spanning decades. In a stable population, the proportion of each age group (cohort, age class, or life stage) at any given time is the same as the proportion of that group over the lifespan of the cohort (e.g., Royce 1984). In an idealized stable population, the dynamic rates of survival/mortality, recruitment, and growth are constant for each age group, and abundances of individual age groups and the population as a whole do fluctuate over time. This idealized stable state is how a population could look if these rates were integrated over many years; however, intrinsic and environmental factors constantly affect populations, causing the ideal state to rarely be seen in the short-term. Dynamic rates and numbers within age groups may vary dramatically between years and at times even drop to zero, for example, when one or more year classes of these long-lived fishes are lost due to reproductive or recruitment failure. Consequently, we cannot define a stable population of bonytail or razorback sucker in OCHs. This is, in part, because stochastic factors affect populations, but also because opportunities for study are limited to a few habitats, and these habitats also must serve both as study opportunities and as production sites, the latter function effectively disrupting any likelihood of stability. Sufficient observational data collected over many years are necessary to estimate dynamic rates and age group abundances or to refine models that describe a self-sustaining population.

Natural systems are considered resilient when they have the capacity to resist and recover from environmental perturbations (e.g., Capdevila et al. 2020). While large populations theoretically are more resilient than smaller ones, other factors play a role in determining their persistence. It is unknown if this intrinsic capacity is com-

promised in these relatively small populations of at most hundreds to thousands of bonytails or razorback suckers that occupy OCHs compared with historically much larger populations in geographically expansive systems. It is virtually certain that these large populations were not homogeneous with respect to demographic rates over space and time. Catastrophic changes can extirpate a naturally occurring or captive population, regardless of size. However, because of the mitigating effects of management, captive populations are not subject to disturbances such as temporary excursions into unfavorable ranges of physicochemical conditions or disruption resulting from stochastically mediated collapse of one or more year classes. Pond-dwelling populations under the OCH concept are expected to vary temporally in abundance and year-class strength, but we predict that maintenance of sufficient (likely hundreds, but as yet unquantified) numbers of genetically diverse (similar to baseline, also unquantified) reproductive adults will impart demographic resilience resulting in recovery and persistence following disturbance. These fish should be maintained in an unspecified number of OCHs to spread the risk of local extirpation and recruitment failure.

Genetic and Demographic Considerations— Maintenance of Genetic Diversity

Maintenance of genetic diversity within populations remains one of the most important considerations when managing imperiled species and was critical in formulation of the OCH concept applied to razorback sucker and bonytail (Minckley et al. 2003). Approaches for characterization of genetic diversity have evolved rapidly in the 20 years since, with newer molecular approaches allowing fine-scaled examination of genetic diversity (reviewed in Turner et al. 2020). These methods now are broadly applied in population genetics studies, including for these 2 species, and are providing essential information for conservation and management applications (see above).

While most early studies of genetic diversity were focused on neutral (i.e., nonadaptive) variation, the importance of adaptive genetic traits was recognized long ago (Lynch 1996, Lynch and Lande 1998) and is even more important today (Teixeira and Huber 2021). Consideration of adaptive variation has largely been theoretical; however, NGS methods have provided evidence for local adaptation of populations (e.g., Hess et

al. 2013, Fitzpatrick et al. 2015) and identification of adaptive genetic variants (e.g., Hohenlohe et al. 2013). These features are important considerations because they can help explain differences in how populations respond to local environmental effects.

Population size is an important consideration when managing imperiled species due to the well-known relationships between number of individuals and adaptive potential (Hoffmann et al. 2017) and between fitness and genetic diversity (Reed and Frankham 2003), with increased population size, distribution, and concomitant levels of genetic variation providing greater resiliency. Concerns over small population size and associated effects of inbreeding in hatcheries are well recognized (Wang et al. 2001, Christie et al. 2014) and have been reported for razorback sucker (Dowling et al. 1996) but not for bonytail.

Concerns about performance of hatchery fish and their impacts on wild populations have also been identified (Heath et al. 2003, Fraser 2008, Osborne et al. 2020). For example, Auld et al. (2021) identified divergence in mating cues between wild and hatchery-reared coho salmon (*Oncorhynchus kisutch*) that could impact wild populations. Also, while hatchery-reared steelhead trout (*Oncorhynchus mykiss*) grow faster in the hatchery than offspring of wild fish, their survival is lower under wild conditions (Blouin et al. 2021). Concerns like these have led to the development of alternative approaches to management of endangered species, such as the OCH plan, and these are useful across species with diverse life history strategies (Osborne et al. 2020).

The OCH plan requires management of populations of bonytail and razorback sucker in isolated ponds and other appropriate habitats, with the goal that fish in OCHs represent and maintain existing diversity in these species. To mitigate impacts of inbreeding in small, finite populations, Minckley et al. (2003) called for frequent transfer of sexually mature recruits among OCHs and from the riverine population to OCHs (and vice versa) to maximize parental contributions to reproduction in the metapopulation. This approach is key to the plan and is feasible for razorback sucker but is not possible for bonytail under current conditions because survival in the river is virtually nil and sufficient repatriates are not available to be collected. Hence, novel strategies, yet to be identified and developed, are required to manage population

genetics of bonytail. Because bonytails mature at a younger age compared to razorback suckers, they may need to be moved more often to obtain similar levels of genetic structuring within and among subpopulations. Note, however, that some differentiation (via genetic drift) is expected among managed populations and is appropriate as long as individuals from these populations retain the ability to interbreed without loss of fitness.

Demographic and genetic processes are strongly linked in population theory (Wright 1969) and in practical implementation of the OCH program. Management actions that address demographic issues like small population size, skewed sex ratio, and high VRS should also positively affect genetic performance. Under ideal conditions, an OCH should harbor as many reproductively capable adults as can be sustained through natural recruitment. The majority of, if not all, reproductive adults should contribute offspring to annual (or generational) cohort(s) of progeny, and an age-structured population should develop after initial stocking. Additionally, a genetically representative progeny should survive to recruit to the population at a rate that sustains adult numbers and genetic diversity. In reality, the number of fish that can be sustained depends on environmental factors like habitat size, water physicochemistry, and temporal variance in habitat quality. Demographic factors are important determinants of population size and levels of genetic diversity maintained in a particular OCH population. Environmental and demographic factors thus are interconnected drivers of population abundance and genetic diversity. Metrics associated with these factors track how close an actual OCH is to the theoretical ideal or desired target conditions.

A key metric that records the combined influences of environmental and demographic factors on genetic diversity is VRS. VRS is inversely proportional to the ratio N_e/N_c or N_b/N_c , where N_c is the number of reproductively capable adults in an OCH. When VRS is relatively low (i.e., near minimum values of VRS that follow a Poisson distribution with $\bar{x} = 1$ offspring per parent per generation), the expected ratio of $N_e/N_c \approx 1$ (Nunney and Elam 1994). The biological interpretation is that most, if not all, adults contribute offspring to the next generation or cohort (in the case of N_b). For annual species, $N_e = N_b$, and contributions to the offspring pool are inclusive and evenly

distributed across parents when VRS is low. Bonytail and razorback sucker are characterized by age-structured populations with overlapping generations where $N_e \neq N_b$ (Waples et al. 2011). When measured across generations, low VRS implies that nearly all reproductively capable adults have contributed to the progeny pool at least once over their lifetimes.

VRS can take on very high values in some fishes, including bonytail and razorback sucker. Both species are long-lived, attain relatively large body sizes, and devote considerable mass and energy to reproduction (Charnov et al. 2001). Females produce thousands of small eggs, and a single mating pair can potentially contribute all or most offspring to the progeny pool. When one or a few mating pairs contribute the majority of offspring that recruit to the next generation, VRS becomes very large, and N_e/N_c is much smaller than 1. This means that the OCH population harbors much less genetic diversity than ideal conditions.

As discussed above, VRS in OCHs was low for bonytail, with the majority of individuals contributing at least one offspring (Osborne et al. 2018). Conversely, standardized VRS in razorback sucker differed greatly across 2 OCHs tested, and across years within 1 OCH. VRS was particularly high among females, and N_b/N_c was correspondingly low (Dowling et al. 2019, unpublished data). Differences in female body size, skipped spawning, costs of previous reproduction, or poor water quality may explain annual VRS (i.e., VRS measured for parents of an annual cohort), but more study is needed to identify specific causes. Whatever the cause, VRS could be lower when measured over a generation (instead of annually) as more females are expected to successfully contribute to the offspring pool over time.

Based on these results, there are 2 key inferences for management. Bonytail appears to perform close to ideally as long as water quality and other environmental conditions are favorable. Razorback sucker, on the other hand, may need multiple years of spawning to ensure that most parents contribute to the generational progeny pool to attain the same level of genetic representation per parent compared with bonytail. It also is likely that skipped spawning, higher-than-expected male mortality, or costs of previous reproduction contribute to the higher VRS observed in razorback sucker. For permanent OCHs, management action should be considered

when N_b/N_c (in bonytail) or N_e/N_c (razorback sucker) is small (e.g., <0.2). Environmental factors such as poor water quality have led to adult mortality and recruitment failure in one or more backwaters. If poor water quality is chronically observed, the OCH should be taken offline to avoid negative effects on the total effective size of the collection of OCHs. When demographic factors are suspected, such as density-dependent changes in fecundity or age- and size-related mortality, then additional stocking and monitoring of reproduction could be conducted.

OCH Stocking, Harvest, Removal, and Transfer

The total number of OCHs required to meet program goals depends on variance in productivity among OCHs and desired level of genetic diversity measured in terms of N_e across all OCHs. To enhance performance of the OCH program as a whole, Minckley et al. (2003) described the need for periodic exchange of adult individuals between OCH populations and the river mainstem, and across OCHs. However, the plan did not specify when, or at what rate, individuals should be exchanged. Transfer of adults from an OCH to the mainstem bolsters riverine populations and prevents exceedance of carrying capacity in donor OCHs. The number and types (i.e., size class, age, or sex) of fish transferred out of an OCH depends on adult densities, year-class strength, and sex ratio. This is a complicated issue as illustrated below. Generally, when densities in a donor OCH are high, more adult fish should be transferred out. For example, adult fish from a dominant cohort may be preferentially selected for transfer to maintain the desired age structure in the donor OCH population or to retain the original, unrelated adults for the long-lived razorback sucker. Likewise, it is prudent to transfer more females out of the population if sex ratio in a donor OCH is strongly female biased. This action restores a 1:1 sex ratio that is optimal for reducing VRS, increasing N_e/N_c , and reducing loss of genetic diversity. Because of low survival, exchange of bonytails with the river is ill-advised. Fish from OCHs could be used to establish new populations, support repatriation efforts into habitats where higher survival is expected, or for other conservation uses. In contrast, the plan as designed should work well for razorback sucker.

Demographic and genetic exchange of individuals across OCHs also achieves important

benefits to overall performance in terms of maintenance and enhancement of genetic diversity. For example, low spawner densities relative to carrying capacity, age structure imbalance, and skewed sex ratios are important reasons, in theory (e.g., Nunney 1996), to transfer adult fish from donor OCHs or the river mainstem to a recipient OCH to achieve sustained recruitment and productivity. Periodic transfer of adults across OCHs simultaneously ameliorates effects of local genetic differentiation and losses of genetic diversity and fitness due to local genetic drift and inbreeding. In theory, as few as 1 adult migrant per generation is sufficient to counter differentiation and inbreeding effects (Whiteley et al. 2015). In practice, values of N_b/N_c (bonytail) or N_e/N_c (razorback sucker) <0.2 indicate that fish should be moved at a higher rate because low values indicate that risks of detrimental genetic effects are correspondingly high (Osborne et al. 2018, Dowling et al. 2019). Actual numbers, rates, and demographic characteristics of transferees should be determined based on regular demographic and genetic monitoring of OCH and river mainstem populations (Tallmon et al. 2004). Monitoring data would be translated to an exchange schedule that meets specific demographic and genetic goals to be developed for a particular OCH, and for the collection of OCHs, in an adaptive management context. Costs of demographic and genetic monitoring and population manipulations should be lowered as initial sampling, analysis, and exchange schedules are refined and optimized.

An Example from Razorback Sucker in Yuma Cove Backwater

An example of applying this approach comes from conservation efforts in Yuma Cove backwater. This habitat was fishless when 200 razorback suckers were stocked in 2013. Since then, a self-sustaining population has been observed and monitored in the backwater. In that time, the site has provided a real-world example of the trade-offs between OCH size, population size, and maintenance of genetic diversity. Through a combination of annual sampling and remote PIT-tag sensing, demographic and genetic trends in a self-sustaining population of razorback sucker were monitored (Dowling et al. 2021). Within a few years of initial stocking, the population had reached more than 400 adults and hundreds to thousands of young-of-year (Dowling et al. 2021). Poor survival of stocked fish in 2014 and 2015,

along with negligible survival of young-of-year fish, suggested resource limitation. An initial attempt to reduce overall biomass was conducted in autumn 2019 when 80 adult and 2877 young-of-year razorback suckers were removed. An additional 100 razorback suckers were stocked the following spring to increase genetic diversity. Twenty-two of the 100 fish stocked were observed alive via remote PIT-tag sensing through 2021, indicating that poststocking survival of these fish was lower than survival observed for the initial stocking in 2013, but higher than stockings in 2014 and 2015—a small but significant indicator that resources were not being overused and thus limiting, at least temporarily.

The apparent overabundance of razorback sucker in Yuma Cove backwater provided an opportunity to implement a transfer of razorback sucker from an OCH and measure its impact on survival in OCH populations. This test removal relied on size classes to identify first-generation recruits from second- or third-generation recruits, targeting the transfer of smaller adults that were likely second- and third-generation fish. It was considered more desirable to retain originally stocked fish and first-generation recruits than more recent (i.e., younger) fish in the OCH because it was assumed these individuals would have greater genetic diversity than offspring of those first-generation recruits. As the population matures, the size classes of different generations will overlap, and it will become impossible to selectively harvest the second-plus generations based on size alone. A general OCH harvest plan will need to rely on both demographic and genetic information and modeling. The plan would need to specify a method to estimate the resultant reduction in overall biomass and the impact those removals would have on survival, spawning biomass, and genetic diversity.

The Off-channel Habitat Concept and Species Recovery

According to the ESA, recovery of a threatened or endangered species is attained when protections of the act are no longer required (16 U.S.C. 1531-1544). For some species, recovery is not achievable, but for others, specific quantitative biological criteria are defined to determine when delisting or downlisting is appropriate. Recovery metrics may be revised and refined as new scientific information is gathered, but revision and refinement may take decades despite available information (e.g., Propst 2020).

Bonytail was listed as endangered in 1980 (USFWS 1980). A recovery plan was published in 1987, updated in 1990, and amended in 2002 (USFWS 1987b, 1990, and 2002a, respectively). The status of bonytail in 1990 was endangered, and it was described as “very rare” with very low to no recruitment. The immediate recovery goal at the time was extinction prevention; longer-term goals leading to down- and delisting, along with appropriate criteria, were not addressed, but 2003 was given as a recovery date for bonytail in the Upper Colorado River basin. That date had come and gone by the time a recovery plan amendment and supplement were published; the species was still endangered, and only a few wild adults remained. The recovery objective, as in all such plans, was down- and delisting. Quantitative criteria required establishment of “genetically and demographically viable, self-sustaining populations.” In 2002, downlisting was considered possible by 2015 and delisting by 2023. No naturally recruiting populations of bonytail exist today so the species remains critically endangered.

Razorback sucker was listed as endangered in 1991 (USFWS 1991a). A recovery plan was published in 1998 (USFWS 1998) and amended in 2002 with addition of quantitative goals and criteria (USFWS 2002b). These plans are summarized here and below. Status of razorback sucker in 1998 was endangered, with the species occurring only in remnant populations. Four years later in 2002, it remained endangered and was described as “in serious jeopardy.” In 1998, downlisting was considered possible by 2010 and delisting as soon as 2020. Little changed and in 2002 dates for downlisting and delisting were estimated as 2020 and 2023, respectively. Both the 1998 and 2002 criteria required populations to be genetically and demographically viable and self-sustaining. No clear statement of the overall status of razorback suckers can be gleaned from the species’ status assessment (USFWS 2018); however, the document suggests qualitative improvement in the condition of existing populations from historical (i.e., 1988) to current (2018) times in both Upper and Lower Colorado River basins. Nonetheless the species inarguably remains biologically imperiled and endangered despite approaching a half century of directed management.

No population of either bonytail or razorback sucker today meets the minimum recovery criterion of being genetically and demographically

viable and self-sustaining. While progress has been made, bonytail is functionally extinct in nature. No “wild” razorback sucker population today is known to be self-sustaining, with most populations dependent upon stocking of hatchery-produced fish. Established populations of predatory and competitive nonnative species continue to thrive and expand, while new species may appear at any time (Propst et al. 2020). In reality, neither bonytail nor razorback sucker is likely to ever meet current recovery criteria, and both species are destined to remain conservation reliant (Goble et al. 2012) until or unless novel and effective management strategies are developed and fully implemented to overcome currently intractable threats. Stocking of hatchery produced, or wild-produced and hatchery-reared (in Lake Mohave) individuals will continue as primary management actions, but no data are available to suggest that these efforts will lead to establishment of self-sustaining populations. Implementation of the OCH concept would result in viable, self-sustaining populations that would not be possible in open waters. These captive populations, although requiring management to maintain diversity and desired age group structure, would provide a continuous source of genetically diverse fish for exchange with natural waters. Although populations of bonytail and razorback sucker have been successfully established in several existing OCHs (Appendices 1, 2), these add relatively few individuals to the total number of fish in the Lower Colorado River. Thus, the need for continued implementation of the OCH concept is urgent. This scenario may be as close as we can get to “recovery” in the foreseeable future.

IMPLEMENTATION OF THE REVISED OFF-CHANNEL HABITAT CONCEPT

While considerable research has been completed on important parameters for a revised version of the OCH plan (reviewed above), there is still much work to do before implementation can positively impact conservation of bonytail and razorback sucker. This includes securing more habitat for use in the OCH plan and acquiring more information on environmental, demographic, and genetic parameters necessary for generation of the metapopulation structure as envisioned by Minckley et al. (2003). Significant logistic hurdles have been largely overcome, foremost of which relate to the availability of

land and water for native fish conservation. The Bureau of Reclamation and its partners through LCR MSCP have examined and evaluated native fish conservation opportunities along the Lower Colorado River corridor for almost 2 decades since inception of the program. That process identified or developed many of the relatively few suitable sites that are available to help meet its goal of creating or protecting and managing 360 acres (146 ha) of OCH for bonytail and razorback sucker and 85 acres (34 ha) for flannelmouth sucker, the former to be isolated but the latter connected to the river and thus unsuitable for uses we envision here. Although only 158 acres (64 ha) had been created as of the end of 2022 (J. Stolberg, Bureau of Reclamation, personal communication), rapid progress is expected in the next few years. Hydrological issues have hampered creation of 60+ acres (24+ ha) at a newly acquired site, but work there is expected to continue in 2025. A new 14-acre (6-ha) pond at Imperial Ponds is to be developed in 2026. Finally, an ambitious project to include 111 acres (45 ha) among 7 disconnected OCHs plus twelve 1-acre (0.4-ha) grow-out ponds and ancillary facilities is slated for construction near Yuma, Arizona, into 2026. Despite some setbacks, when these 3 projects are complete the system will total 343 acres (139 ha) of isolated OCH, with only 17 acres (7 ha) more to be developed.

Achieving habitat goals in terms of surface area of water is only one step toward conservation success. We do not yet have an estimate of OCH carrying capacity in terms of number or biomass of fish per unit area for either bonytail or razorback sucker, and performing experiments to make such a determination has proven difficult for many reasons. Relative carrying capacity is likely to vary across OCHs because of differences among habitats. Knowing carrying capacity could allow an expectation of fish available for exchange with the river channel and among sites, and determination of the size of adult population that could be maintained in each OCH.

Each OCH along the Lower Colorado River is unique, with measurable differences in configuration, surface area, volume, substrate, and aquatic vegetation, and often in water chemistry and abundance and diversity of primary and secondary producers. Thus, there is no opportunity for replication to support traditional common garden experiments. Seasonal and annual

trajectories of these dynamic categories are highly variable and often unpredictable, and these have known effects on fish condition, growth, and reproduction. Complete population failures have resulted from flash flooding at a lakeside site that flushed native fish into Lake Mohave and allowed nonnatives to invade, and from oxygen depletion during hot weather leading to fish kills at Imperial Ponds. There is no mitigation for the former other than alternate site selection away from desert washes, while in the latter instances, extremes of dissolved oxygen and temperature might at least be partially ameliorated by aeration and pumping cooler groundwater into ponds. Unfortunately, there have been instances of potentially lethal environmental conditions resulting from human error even where infrastructure was in place to prevent such situations, suggesting automation of some systems could be beneficial.

Selection of Off-channel Habitats

When the OCH concept was originally proposed, Cibola HLP provided one of the few practical examples of if and how it would work. Since then, there have been several attempts to implement the concept, with varied success. In general, the concept itself has proven sound, but the devil has been in the details. Larger ponds, as predicted by Minckley et al. (2003), have been more difficult to manage, and thus far no self-sustaining population of bonytail or razorback sucker has established in an OCH pond >5 surface acres. Unfortunately, 5 surface acres may not be large enough to contain a population that can maintain both a stable age structure and genetic diversity if such ponds rapidly overpopulate (e.g., 1 to 2 years after establishment).

UCH size is not the only factor that has impacted the effectiveness of larger ponds. Another consideration is the capability to drain a site, but incorporation of this feature appears at odds with the suggested water supply in the OCH concept. A passive water supply, through either subsurface flow or through porous shoreline substrates, is preferable to river water pumped through a filter because the latter cannot reliably preclude direct invasion of nonnatives. However, presence of passive flow can make it difficult or logistically impossible to fully drain and chemically renovate an OCH. Further, the proximity to the water source required to allow passive flow may increase the risk of nonnative introduction through direct transfer. Finally, both

passive flow-through and active filtered pumping do not assist in water temperature amelioration that can be necessary to prevent fish kills during the hottest summer months. Well water is the only secure source that can effectively lower water temperatures and prevent incursion of nonnative species, but this source is dependent on availability of reliable electrical power and on humans to ensure timely water delivery.

Use of design features such as boulders to provide cover for bonytail and gravel spawning substrate for razorback sucker generally have been supported by data collected at OCH research sites. Riprap was used readily by bonytail in Cibola HLP, but boulders emplaced for that purpose in the Imperial Ponds were not as useful because interstitial spaces provided by the thin veneer of rock were too small and quickly filled in with sediment. Recruitment of razorback sucker in Imperial Ponds followed the addition of spawning substrates, and remote PIT scanning on those substrates indicated their use during the known spawning period for razorback sucker. However, recruitment at Imperial Ponds has not reached levels seen in smaller isolated habitats such as Cibola HLP and Yuma Cove backwater. Nonnative fish establishment in Imperial Ponds also likely contributed significantly to failed recruitment, but pond size, complexity, depth, and shoreline profile all may contribute as well. It is difficult to isolate the impact of each factor without the appropriate experiments, which would require a larger number of OCHs than are available at this time.

Development and creation of OCHs is only part of the solution, as not all such habitats work equally well for these purposes. Off-channel ponds along the shores of Lake Mohave (e.g., Appendix 2) in the past have been used to rear bonytail and currently are used to rear razorback sucker to improve their growth, with varying levels of success (Minckley and Thorsen 2007, Ward et al. 2007). As discussed above, there was variation in reproduction and survivorship among ponds and years in studies of reproductive success in bonytail and razorback sucker (Osborne et al. 2018, Dowling et al. 2019). There also was considerable variation in adult survival and reproductive success among the 6 habitats at Imperial Ponds (J. Lantow, Bureau of Reclamation, personal communication). For bonytail, reproduction and recruitment regularly occurred in 2 ponds (2 and 6; photo 4); however, fish kills occurred in 2020 (pond 5) and 2022

(pond 2). Razorback sucker reproduction and recruitment have occurred consistently in Yuma Cove backwater since 2013; however, this has been true for only 1 of the 3 Imperial Ponds occupied by this species, and there were significant fish kills in 2 ponds in summer 2022. It is unclear whether these results would differ if bonytails were stocked into ponds where razorback sucker mortality occurs or vice versa.

Given the high probability that at least some ponds developed for the OCH program will be unsuitable for their intended purpose, it would be valuable to develop tools that allowed identification of OCHs that are likely to be useful before limited resources and effort are invested. This would require detailed study of biotic and abiotic conditions of ponds, allowing comparison of successful and failed ponds to identify conditions that are important determinants of survival, reproduction, and recruitment of these species. Unfortunately, this approach is fraught with difficulty because it likely is a combination of abiotic and biotic features that causes failure of a pond, and it would be difficult to tease out the important factors experimentally. Individual OCH quality is likely to vary from year to year, so this regular monitoring of OCHs is essential.

Development of NGS methods and their application to ecological studies provide a possible solution to this problem because this methodology has been used to characterize biotic communities and their response to abiotic and biotic effects. Several studies have used metabarcoding of water and/or sediment samples. Metabarcoding is an NGS-based method where DNA or RNA is sequenced to identify many specific taxa in a water or sediment sample to assess the role of various organisms as bioindicators of habitat quality. Ankley et al. (2022) used these methods in experimental freshwater mesocosms exposed to an oil spill and monitored the change in the zooplankton community. They found that change in zooplankton species diversity from 3 days prior to an oil spill to 38 days after was a useful indicator for characterizing the ecological response of communities to environmental changes. Emilson et al. (2017) conducted a survey of macroinvertebrates across environmental stream gradients and found that DNA metabarcoding provided equivalent results to traditional morphological approaches and concluded that barcoding of invertebrates could provide a useful tool for broadscale biomonitoring across watersheds.

Biomonitoring of bacterial communities provides an alternative approach for the purposes of assessing the quality of OCHs used to establish self-sustaining populations of bonytail and razorback sucker. Aylagas et al. (2021) assessed the use of DNA-based monitoring of bacterial communities in aquatic sediments, finding that bacteria can function as bioindicators of environmental stressors. Pilgrim et al. (2022) conducted similar characterization of the periphyton-associated bacterial community among localities, identifying bacterial taxa that were associated with changes in total phosphorus and nitrogen concentrations. These results highlight the potential value of bacteria as bioindicators for characterizing environmental differences among OCHs, possibly allowing identification of environmental factors associated with failure and suitable habitats for use in this program.

Filling the Gaps—Demographic and Ecological Information Needed for Implementation

When fully implemented, the program will contain many OCHs integrated into a large metapopulation. Management goals will focus on enhancing demographic and genetic performance in individual OCHs and for the network as a whole. Yet almost everything we know about off-channel habitats comes from study of the relatively few ephemeral and permanent habitats described above. From that information, we used metapopulation theory and intuition to propose how the collective or integrated OCH program will perform, but this proposal will need testing through rigorous monitoring of environmental and demographic factors within each OCH. The integrated OCH program will consist of many individual OCHs, each ostensibly managed to maximize abundance, persistence, and genetic diversity. Metapopulation theory tells us that abundance and genetic diversity are maximized for the program as a whole when OCHs produce roughly equal numbers of recruits per spawner per generation (Nunney 1996), the adult population is stable and age-structured (Nunney 1993), and density is maintained just under carrying capacity through time. In a given year, some habitats will be more productive than others, and some might fail entirely with local extirpation. High variance in productivity and high extirpation risk in space and time diminish the number of adults and genetic diversity in the program as a whole (e.g., Vucetich et al. 1997). Therefore, identifying, monitoring, and mitigating

ecological factors that drive differences in productivity among individual OCHs should ultimately lead to more adult fish, higher genetic diversity, and better recruitment for the program as a whole. Individual OCHs that consistently perform poorly should be removed from the program, preferably in concert with introduction of consistently productive OCHs into the program. Generally speaking, the number of OCHs needed to meet program goals must be increased when variability in productivity and extirpation risk across individual OCHs is high.

While population genetic parameters are relatively easy to quantify, monitoring of demographic factors among adults can only be done readily through remote PIT-tag scanning technology. This approach contributes little to our knowledge of direct sources of larval and juvenile fish mortality and other ecological factors that underlie recruitment success in individual OCHs. We do not know the nature of density dependence and its influence over the entire life cycle of these fishes, particularly on recruitment success. We also do not know how natural and anthropogenic selection operate in individual OCHs. Parentage-based tagging approaches, such as those employed in the management of Pacific salmon (e.g., Steele et al. 2019), should be employed to monitor VRS and to test for nonrandom survivorship of particular genotypes in individual OCHs, especially when compared genetically to natural and hatchery populations throughout the Colorado River basin.

Preliminary observations at Imperial Ponds and Yuma Cove backwater on Lake Mohave raised a number of immediate issues and questions, including some of those identified in Minckley et al. (2003), that should be addressed to maximize the productivity of OCHs. These include (1) cannibalism of eggs/larvae in bonytail and what habitat modifications can be made to ponds to reduce this impact, (2) density-dependent effects in confined backwater habitats and how these function in determining carrying capacity, (3) variation in function among OCHs (e.g., differences in bonytail and razorback sucker reproduction between sites at Imperial Ponds, razorback sucker in ephemeral ponds), and (4) periodic loss of entire populations due to water-quality issues in OCHs both at Lake Mohave and Imperial Ponds. The last requires either redundancy in the number of ponds so system-wide numbers of adults and progeny can be maintained despite such losses, or a need for

TABLE 2. Recovery goals for bonytail (USFWS 1987b, 1990, 2002a).

Plan year	Primary goal/objective	Downlisting criteria	Delisting criteria	Priority actions needed
1987	Prevent extinction	Four viable, self-sustaining populations	Six viable, self-sustaining populations in protected habitats	Compile historical data; improve identification; sample, monitor, assess impacts; study hybridization; select stocking sites
1990	Prevent extinction	Not addressed	Pending extinction threat removal	Develop genetically diverse captive population; study life history and habitat; resolve taxonomy; develop quantitative goals
2002	Down- and delisting	Over 5 years: more than 4400 adults in Green River plus an additional upper basin population; a genetic refuge in the lower basin; 2 populations in the lower basin, each more than 4400 adults; threat management	Delisting plus over 3 years: 4400 adults in Green River plus an additional upper basin population; a genetic refuge in the lower basin; 2 populations in the lower basin, each more than 4400 adults; threat management	Stock to reestablish populations; maintain genetic refugia; legally protect habitat; provide passage over barriers; plus 10 other actions

TABLE 3. Recovery goals for razorback sucker (USFWS 1998, 2002b).

Plan year	Primary goals/objectives	Downlisting criteria	Delisting criteria	Priority actions needed
1998	Short-term: prevent extinction; long-term: down- and delisting	More than 50,000 adults in L. Mohave for 5 years; more than 5000 adults in the Yampa-Green rivers for 5 years; management practices; 2000 adults in lower Green River; 2 new populations; natural recruitment	2 additional populations (1 each in the upper and lower basins); 8 populations secure from threat and legally protected	Protected genetic refugia; habitat restoration; control nonnative fishes; augmentation stocking; population and habitat monitoring
2002	Down- and delisting	Over 5 years: more than 5800 adults in Green River plus an additional upper basin population; genetic refuge in L. Mohave; 2 lower basin populations of more than 5800 adults; threat management	Delisting plus over 3 years: 5800 adults in Green River plus an additional upper basin population; genetic refuge in L. Mohave; 2 lower basin populations of 5800 adults; threat management	Stock to reestablish populations; maintain genetic variability in L. Mohave; legally protect habitat; provide passage over barriers; plus 10 other actions

reliable surveillance at least of temperature and dissolved oxygen so that losses can be mitigated or eliminated entirely.

Evaluation of Goals and Outcomes

LCR MSCP conservation goals and objectives are quantitatively defined as number of surface acres of water and number of fish stocked. These are easily measured and tracked, and the program appears on a trajectory that will arrive at its designated endpoints in a timely manner (e.g., LCR MSCP 2021). Recovery goals and objectives quantified as population criteria (Tables 2, 3) are another matter entirely. Assessing progress requires estimation of population census size, and progress toward attainment has been slow. Goals promulgated in the earliest recovery plans for bonytail and razorback sucker were qualitative and called for prevention of species extinction. Bonytail has been maintained in captivity and there is an on-going hatchery-based stocking program that has failed to establish any wild population; thus, quantitative recovery criteria are meaningless for this species. The Lake Mohave Native Fishes Work Group set its sights on restoring a population of 50,000 adult razorback suckers in Lake Mohave that was genetically representative of the historical population in that reservoir (Mueller 1995, Dowling et al. 1996), and that goal was among the downlisting criteria in the 1998 but not the 2002 recovery plan. Intensive effort over more than 3 decades has succeeded in maintaining a population of about 10% of that goal (Marsh et al. 2015, Saucier et al. 2022), and genetic diversity thus far has been maintained (Dowling et al. 2014). Establishment of one or more additional populations of similar size in the Lower Colorado River downstream of Lake Mohave would be a serendipitous outcome of the LSC-MSCP, but those fish currently are derived from hatchery production and do not have the same high level of genetic diversity as the Mohave stock. Given the virtually limitless availability of larvae from Lake Mohave, the program might consider exclusive use of these fish in lieu of artificial propagation.

Assessing success of an OCH-based conservation program as envisioned here also requires quantitative studies to estimate key population parameters discussed earlier (at the least numerical abundance and recruitment) and genetic characteristics that can be compared with histor-

ical information and with current goals and objectives. Genetic tools have been used to characterize several OCH populations and have provided insight into important genetic and demographic parameters that are essential for evaluating the success of individual OCHs (Dowling et al. 2014, 2019, 2021, Osborne et al. 2018). Especially important are estimates of N_e and N_b generated for both bonytail (Hedrick et al. 2000, Osborne et al. 2018) and razorback sucker (Turner et al. 2007, Dowling et al. 2014), and estimates of relatedness (Appendix 3), which have yet to be generated. This information is especially valuable in a long-term monitoring situation where population census data also are available (Dowling et al. 2014) and will allow assessment of the impact of management actions on genetic and demographic parameters.

A critical aspect of the OCH program is adaptive management, or a periodic assessment that (1) provides data to compare with goals and objectives as described earlier and (2) serves also as the feedback mechanism for positive change. For example, NGS approaches have been developed to provide more accurate estimates of genetic diversity parameters (see Appendix 3), providing information necessary for the maintenance of self-sustaining populations that are viable in the long term. Such metrics will be used to guide management actions, such as harvest of individuals from ponds as they become overpopulated and movement of individuals among OCHs to maintain genetic diversity and decrease relatedness of individuals.

CONCLUSIONS

Minckley et al. (2003) developed the OCH program as a means to conserve the endangered large river fishes of the Colorado River, especially 2 of the species that once were abundant in the lower basin. Twenty years have passed since the publication of OCH program. Some advances have been made, yet much work still needs to be done. Unfortunately, nonnative species, limited habitat, climate change, and other threats that were in place in 2003 persist and may be worse at the present time. Survival of bonytail in the river proper is poor (e.g., Mueller 2006, Bestgen et al. 2008), and there is no evidence of long-term persistence anywhere; therefore, this species has been relegated to backwater populations. Razorback sucker exists in

mainstem habitats; however, survival has been reduced and recruitment is nil, requiring the stocking programs to maintain its presence in virtually all places where it persists. Progress toward recovery requires reduction or elimination of threats and constraints to survival and recruitment. Until such progress can be attained, conservation of these species necessitates programs like those proposed here and by Minckley et al. (2003).

Given these circumstances, it is necessary to implement programs that allow these species to persist in a state that is amenable to further conservation efforts, maintaining them in the wild for at least some part, if not all, of their lives (reviewed in Osborne et al. 2020). The genetic legacy of bonytail is represented today by hatchery stocks derived from artificial propagation of a small number of individuals harvested from the wild more than 30 years ago (Hedrick et al. 2000) and without subsequent augmentation. Ironically, bonytail does exceptionally well in hatchery and isolated pond settings absent nonnative fishes, but the species fails elsewhere. It is unclear whether reduced genetic diversity has any impact on current efforts involving bonytail or whether only short-term environmental effects are operative in these situations.

Razorback sucker fares differently because the repatriated population that now occupies Lake Mohave retains the diversity of its predecessor (Dowling et al. 2021), and this is due entirely to success of the innovative management program practiced in that reservoir (Mueller 1995, Bestgen et al. 2020); however, Lake Mohave currently is the only place where this legacy is preserved. There, razorback sucker larvae can be collected in large numbers in mainstem habitats, reared in hatcheries and isolated habitats, and released at a size with measurable survival when the fish are placed back into the wild. Poststocking survival is positively related to size at release; thus, efforts are ongoing to grow fish as large as possible within constraints of time, resources, and hatchery operations. This approach has been used at Lake Mohave since the mid-1990s (reviewed in Marsh et al. 2015) and has successfully maintained much of the genetic variation that was found in the original population (Dowling et al. 2014, 2021).

The next step is to utilize predator-free OCHs to retain or increase the “wildness” of these endangered species, enhancing their probability

of persistence. The OCH plan seeks to achieve both demographic goals (increase population size, improve survivorship, preserve life history traits) and genetic targets (maintain historical levels of diversity by maximizing parental contributions [i.e., N_e]). The plan whereby adults reproduce and recruit in predator-free habitats is beneficial on many levels because natural spawning and other behaviors are maintained (e.g., foraging, response to spawning cues, mate choice, spawning site selection) and there is potential for contribution from a far larger number of parents than is possible in a traditional hatchery setting. Exchange of individuals among OCHs and mainstem populations (where possible) will more closely emulate the former connectedness of metapopulations of these species, moving us closer to stability and adaptability of these once widely distributed species.

While the OCH plan was developed for fishes of the Lower Colorado River, its success for bonytail and razorback sucker could serve as a template for other imperiled species where recruitment failure is a primary impediment to species recovery (e.g., Osborne et al. 2020). The OCH plan is most amenable to species that share key life history traits with bonytail and razorback sucker, including high fecundity and absence of parental care, and where spawning sites are known and accessible.

Further, OCHs could provide important indirect benefits to the river ecosystem. For example, shallow, lentic habitats often are highly productive and contribute energy and materials to riparian ecosystems through emergent insects. They also provide some habitat heterogeneity to the river system as a whole and support a different community of aquatic flora and fauna than the mainstem, and thus contribute to overall ecosystem diversity.

Importance of Redundancy

For the OCH program to be successfully implemented and to continue to preserve the genetic integrity of razorback sucker, redundancy must be an integral part of the system. This is important because experience has shown that individual OCHs are subject to failure (i.e., loss of contained fish) for several reasons, and situations can easily be envisioned in which multiple OCHs might be lost at the same time and for similar reasons. Redundancy, therefore, must extend beyond simply having multiple OCHs at

individual conservations areas (e.g., there are 6 [soon to be 7] sites at Imperial Ponds, and 2 areas under development are to have 4 and 7 sites, respectively; J. Stolberg, Bureau of Reclamation, personal communication). The LCR MSCP planning area extends along the Lower Colorado River corridor approximately 450 miles (724 km) from Separation Rapids at the head of Lake Mead to the international boundary with Mexico, and along this path the stream flows through reaches of variable geology, hydrology, local climate, plant and animal community type, land use, and political jurisdiction. To achieve redundancy, OCH sites also must be distributed throughout the system to reflect this inherent variability, both to protect existing genetic diversity and to provide a system in which selection can proceed along a natural trajectory.

Conservation Reliance – The Final Outcome

Aggressive management of razorback sucker in the Lower Colorado River began more than 4 decades ago in the early 1980s (Minckley et al. 1991). In that time, the Lake Mohave population of razorback sucker has dwindled from tens of thousands of wild fish to about 5000 repatriates, and bonytail has become functionally extinct in the wild. Neither species recruits naturally in historical habitats, and persistence of both is entirely dependent on stocking. In context, the LCR MSCP has a finite life of 50 years, about the life span of a razorback sucker and twice that of a bonytail. The program commenced in 2005 and will end in 2055 (at this writing into year 18 with 32 remaining). Valuable lessons have been learned about these fishes, and there is enough understanding of their respective life histories to enable managers to perpetuate them, at least in the relatively short term. Unfortunately, the primary factors responsible for their imperiled status—presence of predatory nonnative fishes and habitat alteration—remain in full force with no likelihood of amelioration. It therefore is clear that bonytail and razorback sucker will be dependent on human intervention for their continued survival as conservation-reliant species. Unlike the LCR MSCP, the ESA will be in effect in perpetuity unless eliminated by Congress, and bonytail and razorback sucker will continue to have protections and conservation management under its authority. And, as the ESA was the root of the legal requirement under which the current LCR MSCP was created, so may a future iteration of the MSCP be created under which endan-

gered species conservation may be continued until recovery is achieved, politically if not biologically, or the 2 species wink out of existence.

ACKNOWLEDGMENTS

The authors owe a debt of gratitude to the late W.L. Minckley, who provided insight and inspiration during the development of the off-channel habitat program. This project would not exist without his years of experience with fishes and ecosystems of the desert Southwest, his intellectual contribution, and his dogged determination to conserve the resources of the region. The general ideas represented here were largely available in some form at the start of the program and have been further modified and developed over the past 2 decades in conversations among the authors and with colleagues too numerous to mention individually. Included are T. Burke (Bureau of Reclamation, retired) and J. Lantow (Bureau of Reclamation), C.O. Minckley (USFWS, retired) and G.A. Mueller (USGS, retired). Likewise, many individuals have been involved in sampling and data collection and management over the last 2 decades. We especially thank C.A. Pacey, who provided current bonytail and razorback sucker stocking data, and K. Shollenberger, who created the map (both Marsh & Associates, LLC). E. Loomis and J. Stolberg (Bureau of Reclamation) reviewed backwater use and stocking data. J. Sosin and A. Wicks (WSU) provided comments on the manuscript. Three anonymous referees provided an abundance of insightful and helpful suggestions regarding the initial submission. Any errors or omissions are the responsibility of the authors. This research has been generously funded by the LCR MSCP, Bureau of Reclamation, and USFWS. Publication costs were waived by a grant from the *Western North American Naturalist*.

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Received 28 February 2023

Revised 25 August 2023

Accepted 22 September 2023

Published online 15 May 2024

APPENDIX 1. Off-channel habitats.

The 3 sites described below figured prominently in early development and initial implementation of the off-channel habitat (OCH) concept. They represent a range of habitat types and formation dynamics available along the Lower Colorado River corridor and were important sites of research and experimental investigations that served to guide design, construction, and modification of OCHs.

Appendix 1.1. Cibola High Levee Pond (HLP). An example of a seminatural off-channel habitat adjacent to the Lower Colorado River (Photo 3).

LOCATION.—This site is near river mile 96 (km 154.5) upstream from the Southerly International Boundary, adjacent to and west of the Colorado River channel on the Arizona-California border south of Blythe, California (Fig. 1). It is part of USFWS Cibola National Wildlife Refuge.

ORIGIN.—HLP is a cutoff oxbow of the river created in the 1960s when the river was channelized and lined with coarse (boulder-size) riprap.

DIMENSIONS.—4.9 surface acres (2 ha), dimensions approximately 1150 × 328 ft (351 × 100 m), maximum depth to 10+ ft (3+ m). Volume 6.6 million gallons (25,000 m³).

SUBSTRATE.—Riprap (east and west shorelines), limited gravels at boat ramp, silt, organic debris, brush, root balls.

WATER SOURCE.—Percolation through the levee/berm from the adjacent river channel; pond elevation varies with river stage.

VEGETATION.—Spiny naiad *Najas marina*, sago pondweed *Potamogeton pectinatus*, cattail *Typha*, salt cedar *Tamarix*, mesquite *Prosopis* sp. and coyote willow *Salix exigua*.

MANAGEMENT AND NOTES.—Chemically treated in 1993 to remove nonnative fishes and stocked with juvenile bonytail and razorback sucker. Periodic removals and additions but no system-wide manipulations.

Appendix 1.2. Imperial Ponds. An example of a wholly artificial, off-channel pond complex (Photo 4).

LOCATION.—This site is near river mile 59 (km 95) upstream from the International Boundary, adjacent to and east of the Colorado River channel northeast of Yuma, Arizona (Fig. 1). It is part of USFWS Imperial National Wildlife Refuge.

ORIGIN.—The Imperial Ponds site originally comprised 4 artificial ponds that were set aside for native fish and waterfowl and named the “DU” or “Ducks Unlimited” ponds. The site was reworked in 2007 and Imperial Ponds today comprise 6 artificial ponds.

DIMENSIONS.—The 4 original ponds totaled 45 surface acres (18 ha). The 6 new ponds total 74.2 acres (30 ha), and design surface areas were Pond 1 (8.8 acres [3.6 ha]), Pond 2 (12.5 acres [5.1 ha]), Pond 3 (14.2 acres [5.7 ha]), Pond 4 (8.6 acres [3.5 ha]), Pond 5 (21.3 acres [8.6 ha]), and Pond 6 (8.8 acres [3.6 ha]). Pond depths were all near 12 ft (3.7 m) deep. Hummocks were included in ponds 3, 4, and 6.

SUBSTRATE.—Riprap (west shoreline of Pond 1), limited gravels at boat ramps and added as spawning substrate, local soils and silt.

WATER SOURCE.—Currently individual groundwater pumps for each of the 6 ponds. An initial system that used river water passed through a wedge-wire screen failed to remove nonnative fish eggs and larvae. The river water replacement system consisted of a single groundwater pump that delivered water individually to each pond via a manifold, but it proved to have inadequate capacity to maintain acceptable physicochemical conditions in the ponds, which suffered in summer from low dissolved oxygen and high temperature.

VEGETATION.—Spiny naiad *Najas marina*, sago pondweed *Potamogeton pectinatus*, cattail *Typha* (extensive in some ponds), salt cedar *Tamarix*, mesquite *Prosopis*.

MANAGEMENT AND NOTES.—Chemically treated to remove nonnative fishes and stocked with juvenile bonytail and razorback sucker. Periodic fish removals and additions but no system-wide manipulations. Ponds 1, 3, and 4 were stocked with razorback sucker; ponds 2, 5, and 6 were stocked with bonytail. Terrestrial shorelines planted with native vegetation.

Appendix 1.3. Yuma Cove Backwater (Yuma). An example of a natural off-channel habitat on a reservoir (Photo 5).

LOCATION.—This site is adjacent to the Lake Mohave (Colorado River) shoreline on the Arizona side of the reservoir at river mile 24.5 (km 39.4) upstream from Davis Dam (Fig. 1). It is on U.S.

National Park Service Lake Mead National Recreation Area.

ORIGIN.—Yuma is a natural site created by accumulation of wind-driven coarse gravel and cobble to form a semipermanent berm that isolates Yuma Cove from the lake. The berm was subject to breaching during periods of high westerly winds when lake elevations were at or near the 647-ft maximum and then reformed when conditions changed. The berm was artificially enhanced using heavy equipment in 1999 and most recently during low lake elevation in autumn 2010.

DIMENSIONS.—Surface area 2.6 acres (1.1 ha) and depth 13 ft (4 m) at lake elevation of 643 ft (196 m), both varying with lake elevation.

SUBSTRATE.—Coarse gravel and cobble, sand, silt, organic debris.

WATER SOURCE.—Passive percolation through the gravel berm from the adjacent reservoir; backwater elevation varies with reservoir elevation.

VEGETATION.—Spiny naiad *Najas marina*, sago pondweed *Potamogeton pectinatus*, salt cedar *Tamarix*, mesquite *Prosopis*, cottonwood *Populus fremontii*.

MANAGEMENT AND NOTES.—Direct management has included augmentation of the berm (above), placement of a fabric mat to inhibit growth of nuisance emergent macrophytes (later removed), and deployment of a wind-driven pump to circulate water and maintain suitable water physicochemistry (i.e., dissolved oxygen and temperature). Yuma has been used to examine razorback sucker population dynamics since the early 1990s. Stockings were of ripe adults, fertilized eggs, larvae, and juveniles followed by population and genetic monitoring (Marsh et al. 2015).

APPENDIX 2. Lake Mohave backwaters.

When Davis Dam was impounded and Lake Mohave filled in the early 1950s, the new reservoir inundated the surrounding landscape including ephemeral desert washes where coves or embayments were created. In some such places, shoreline sediments are redistributed by wind-driven waves and deposited as bars or berms across the cove mouth to isolate the site from the lake proper and create a backwater. As the reservoir elevation fluctuates across its nominal 15-ft (4.6-m) vertical range, these backwaters may alternately fill and desiccate; maximum elevation is in mid-May and minimum in October. Most sites are ephemeral, but a few are perennial. These off-channel habitats or lakeside ponds have been utilized since 1992 for a suite of experimental and research purposes, and for bonytail and razorback sucker rearing and grow-out. Currently, habitats are stocked with juvenile fish as the reservoir fills in late January through mid-March and then harvested in autumn as the water level declines. In the descriptions below, the 3 sites in

italics indicate those in use as grow-out pods as of 2023.

Arizona Juvenile (AJ)

LOCATION.—Lake Mohave shoreline in Arizona, 15.5 miles (25 km) upstream from Davis Dam (Fig. 1). It is in Lake Mead National Recreation Area.

ORIGIN.—A shoreline cove where wave action redistributes lake substrates to form a natural berm.

DIMENSIONS.—Surface area and depth vary with lake elevation; 7.5 ft (2.3 m) deep, 0.3 acres (0.1 ha) at lake elevation 643 ft (196 m).

SUBSTRATE.—Mostly silt and detritus, coarser materials including cobble at berm.

WATER SOURCE.—Water exchange with Lake Mohave through a porous berm. The site is ephemeral (dries in autumn; min. lake elevation is in October).

VEGETATION.—Seasonal rooted aquatic macrophytes (*Najas*, *Potamogeton*), summertime floating algal mats. Chronic cattail (*Typha*) infestations, bulrush (*Scirpus*), and coyote willow (*Salix exigua*).

MANAGEMENT AND NOTES.—The berm has been destroyed by monsoon-related flood flows and restored twice since 2015, allowing resident fish to escape into the reservoir and nonnatives to occupy the site. Periodic control of nuisance cattail and floating algal mats has been practiced. AJ has been stocked with razorback sucker for study and grow-out and was a primary site for population genetics research.

Dandy

LOCATION.—Lake Mohave shoreline in Nevada, 15.5 miles upstream from Davis Dam (Fig. 1). It is in Lake Mead National Recreation Area.

ORIGIN.—A shoreline cove where wave action redistributes lake substrates to form a natural berm.

DIMENSIONS.—Surface area and depth vary with lake elevation; 8.3 ft (2.5 m) deep, 0.4 acres (0.2 ha) at 643 ft (196 m) lake elevation.

SUBSTRATE.—Silt and detritus with small areas of lake-derived cobble; coarser materials including cobble at berm.

WATER SOURCE.—Water exchange with Lake Mohave through a porous berm. The site is ephemeral (dries in autumn; min. lake elevation is in October).

VEGETATION.—Seasonal rooted aquatic macrophytes (*Najas*, *Potamogeton*), summertime floating algal mats. Cattail (*Typha*), bulrush (*Scirpus*), and coyote willow (*Salix exigua*).

MANAGEMENT AND NOTES.—Periodic control of nuisance floating algal mats has been necessary. Dandy has been stocked with razorback sucker for study and grow-out and was a primary site for population genetics research.

Nevada Egg

LOCATION.—Lake Mohave shoreline in Nevada, 15.8 miles upstream from Davis Dam (Fig. 1). It is in Lake Mead National Recreation Area.

ORIGIN.—A shoreline cove where wave action redistributes lake substrates to form a natural berm.

DIMENSIONS.—Surface area and depth vary with lake elevation; 6.4 ft (1.9 m) deep, 0.3 acres (0.1 ha) at 643 ft (196 m) lake elevation.

SUBSTRATE.—Mostly silt and detritus, coarser materials including cobble at berm.

WATER SOURCE.—Water exchange with Lake Mohave through a porous berm. The site is ephemeral (dries in autumn; minimum lake elevation is in October).

VEGETATION.—Seasonal rooted aquatic macrophytes (*Najas*, *Potamogeton*), summertime floating algal mats. Cattail (*Typha*), bulrush (*Scirpus*), and coyote willow (*Salix exigua*).

MANAGEMENT AND NOTES.—Nevada Egg has been stocked with bonytail for study and grow-out and was a primary site for population genetics research.

Nevada Larvae

LOCATION.—Lake Mohave shoreline in Nevada, 14.8 miles upstream from Davis Dam (Fig. 1). It is in Lake Mead National Recreation Area.

ORIGIN.—A shoreline cove where wave action redistributes lake substrates to form a natural berm.

DIMENSIONS.—Surface area and depth vary with lake elevation; 10.5 ft (3.2 m) deep, 0.2 acres (0.1 ha) at 643 ft (196 m) lake elevation.

SUBSTRATE.—Mostly silt and detritus, coarser materials including cobble at berm.

WATER SOURCE.—Water exchange with Lake Mohave through a porous berm. The site is ephemeral (dries in autumn; minimum lake elevation is in October).

VEGETATION.—Seasonal rooted aquatic macrophytes (*Najas*, *Potamogeton*), summertime floating algal mats. Cattail (*Typha*), bulrush (*Scirpus*), and coyote willow (*Salix exigua*).

MANAGEMENT AND NOTES.—Nevada Larvae was stocked in 2014 with 160 adult bonytail that died in a fish kill shortly after stocking, and the site has not been used for native fishes since.

North Chemehueve

LOCATION.—Lake Mohave shoreline in Nevada, 12.5 miles upstream from Davis Dam (Fig. 1). It is in Lake Mead National Recreation Area.

ORIGIN.—A shoreline cove where wave action redistributes lake substrates to form a natural berm.

DIMENSIONS.—Surface area and depth vary with lake elevation; 10.3 ft (3.1 m) deep, 0.5 acres (0.2 ha) at 643 ft (196 m) lake elevation.

SUBSTRATE.—Mostly silt and detritus, coarser materials including cobble at berm.

WATER SOURCE.—Water exchange with Lake Mohave through a porous berm. The site is permanent.

VEGETATION.—Seasonal rooted aquatic macrophytes (*Najas*, *Potamogeton*), summertime floating algal mats. Cattail (*Typha*), bulrush (*Scirpus*), and coyote willow (*Salix exigua*).

MANAGEMENT AND NOTES.—North Chemehueve has been stocked with razorback sucker for study and grow-out.

North Nine-Mile

LOCATION.—Lake Mohave shoreline in Nevada, 16.25 miles upstream from Davis Dam (Fig. 1). It is in Lake Mead National Recreation Area.

ORIGIN.—A shoreline cove where wave action redistributes lake substrates to form a natural berm.

DIMENSIONS.—Surface area and depth vary with lake elevation; 5.5 ft (1.7 m) deep, 0.3 acres (0.1 ha) at 643 ft (196 m) lake elevation.

SUBSTRATE.—Mostly silt and detritus, coarser materials including cobble at berm.

WATER SOURCE.—Water exchange with Lake Mohave through a porous berm. The site is ephemeral (dries in autumn; minimum lake elevation is in October).

VEGETATION.—Seasonal rooted aquatic macrophytes (*Najas*, *Potamogeton*), summertime floating algal mats. Cattail (*Typha*), bulrush (*Scirpus*), and coyote willow (*Salix exigua*).

MANAGEMENT AND NOTES.—Lakeside margin of the site deepened mechanically in 2015. North Nine-mile has been stocked with bonytail for study and grow-out and was a primary site for population genetics research. It has not been used for native fishes since 2016.

Willow

LOCATION.—Lake Mohave shoreline in Nevada, 16 miles upstream from Davis Dam (Fig. 1). It is in Lake Mead National Recreation Area.

ORIGIN.—A shoreline cove where wave action redistributes lake substrates to form a natural berm.

DIMENSIONS.—Surface area and depth vary with lake elevation; 6.7 ft (2.0 m) deep, 0.4 acres (0.2 ha) at 643 ft (196 m) lake elevation.

SUBSTRATE.—Mostly silt and detritus, coarser materials including cobble at berm.

WATER SOURCE.—Water exchange with Lake Mohave through a porous berm. The site is ephemeral (dries in autumn; minimum lake elevation is in October).

VEGETATION.—Seasonal rooted aquatic macrophytes (*Najas*, *Potamogeton*), summertime floating algal mats. Cattail (*Typha*), bulrush (*Scirpus*), and coyote willow (*Salix exigua*).

MANAGEMENT AND NOTES.—Willow has been stocked with razorback sucker for study and grow-out.

APPENDIX 3. Metrics to assess genetic and demographic parameters in off-channel habitats.

Genetic monitoring of populations has been achieved using many different statistical estimators that characterize various aspects of genetic variation (reviewed in Allendorf et al. 2022). Most

molecular markers currently in use (reviewed in Turner et al. 2020) identify genetic variants that do not respond to natural selection (i.e., specific variants that do not provide a benefit to the individual that has them); therefore, they provide an assessment of the impact of genetic drift on the population of interest. Genetic drift is the random change in allele frequencies from generation to generation because of sampling error associated with reproduction in a finite population. The impact of drift is related to population size, with greater change in allele frequencies from one generation to the next and more rapid loss of alleles over time occurring in smaller populations. In addition, fewer alleles are expected in smaller populations due to greater fluctuations in allele frequencies and more rapid loss of alleles.

Many different metrics are used to characterize and monitor population and individual genetic diversity like those found in off-channel habitats (see below). Two commonly used measures of genetic variation are *allelic richness* (Petit et al. 1998) and *heterozygosity* (Nei 1987). *Observed heterozygosity* (H_{obs}) is based on the proportion of heterozygous individuals observed in a sample, while *expected heterozygosity* (H_{exp}) can be calculated from allele frequencies, assuming the sample came from an ideal population (i.e., a population in Hardy–Weinberg equilibrium—random mating, equal sex ratios, etc.). These estimates of heterozygosity allow for calcula-

tion of the *inbreeding fixation index* (F_{IS}) (Wright 1950), a measure of the impact of nonrandom mating within a population that is associated with small population size. Estimates of heterozygosity can also be standardized across loci (*standardized multilocus heterozygosity*, *sMLH*; Coltman et al. 1999). This approach standardizes heterozygosity when not all loci are typed in each individual, which often occurs when NGS approaches are applied. Another important metric used to characterize small populations like those found in off-channel habitats is *internal relatedness* (Amos et al. 2001), with higher values being a signal of increased breeding between related individuals.

Levels of genetic variation can be used to estimate *effective population size* (N_e) or the *effective number of breeders* (N_b). Typically, N_e and N_b provide measures of population size for an ideal population with the same level of genetic variation as found in the sample population. Effective population sizes typically are smaller than *census population size* (N_c) due to reduced genetic contribution of some individuals in the population, in other words, when the population experiences high *variance in reproductive success* (VRS) among individuals. N_e can be estimated in several different ways depending upon one's needs, and several metrics typically used for purposes such as ours are provided in the listing below.

Metric	Description
Allelic richness	Total number of alleles/haplotypes in a population corrected by rarefaction to account for differences in sample size among collections.
Heterozygosity (H_{exp} and H_{obs})	Presence of different alleles at one or more loci on homologous chromosomes. Proportion of heterozygous individuals for a locus in a population. These can be either based on observations (H_{obs}) or estimated from the data (H_{exp}), assuming the sample came from an ideal population.
Fixation index (F_{IS})	Inbreeding at the population level with respect to nonrandom mating within demes. Measures heterozygote deficiency or excess per locus. $F_{IS} = (H_{exp} - H_{obs})/H_{exp}$
Standardized multilocus heterozygosity (sMLH)	Mean heterozygosity across all genotyped loci divided by mean heterozygosity at loci genotyped in the population (i.e., temporal sample). Standardizes heterozygosity when not all loci are typed for each individual.
Census population size (N_c)	The number of individuals estimated to be in a population. This can be obtained through mark-recapture methods or PIT-tag remote scanning.
Genetic effective population size (N_e , N_b)	Genetic effective size of a breeding population under idealized conditions meeting the assumptions of Hardy–Weinberg (i.e., equal sex ratio, random mating). In species with long life, it is possible to sample specific year classes to estimate the effective population size of that specific cohort, which would reflect the number of breeding individuals (N_b) producing that cohort.
Variance in reproductive success (VRS)	The variability in genetic contribution of individuals to reproduction (measured by number of progeny per adult). VRS may be estimated over the life of a cohort, annually, or on another time basis.
Internal relatedness (IR)	$IR = (2H - \sum f_i^2) / (2N - \sum f_i^2)$, where H is the number of homozygous loci, N is the number of loci, and f_i is the frequency of the i th allele in the genotype.